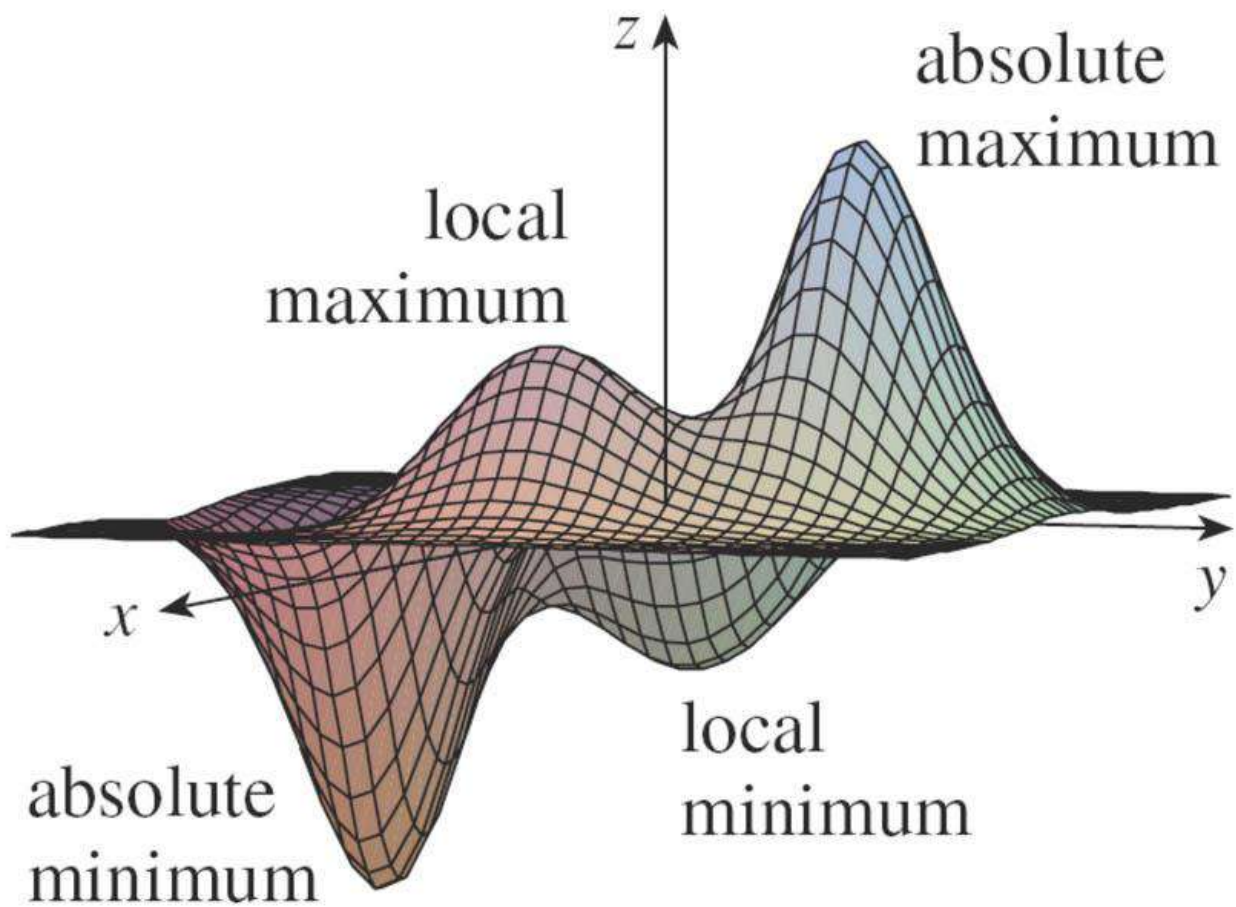

Calculus Summary



Content:

1. Introduction

2. Branches of Calculus

3. Function and Its Types

- **3.1 Scalar-Valued Function**
- **3.2 Vector-Valued Function**

4. Contour Plot

5. Vector Field

6. Gradient Vector Field

7. Function Transformation

8. Limits

9. Intuition of Gradient

10. Differentiability & Continuity

11. Chain Rule

- **11.1 Single-Variable Chain Rule**
- **11.2 Multivariate Chain Rule**

12. Partial Derivatives

13. Directional Derivative

14. Intuition of Gradient Vector

15. Gradient Vector Field

16. Hessian Matrix

17. Jacobian Matrix

18. Numerical Solution

- **18.1 Newton–Raphson Method**

19. Taylor Series Approximation

20. Exponentially Weighted Moving Average (EWMA)

Calculus

→ Is the mathematical study of continuous change.

* Calculus can be considered as a set of tools for analyzing the relationship between functions & their inputs.
"Rate of Change"

Calculus has Two Branches:

Differential

Concern:

- Instantaneous Rate of change and
- slopes of curves.

Integral

Concern:

- Accumulation of quantities and areas under or between curves



* Multi Variable Calculus:

→ Calculus in higher Dimensions.

↳

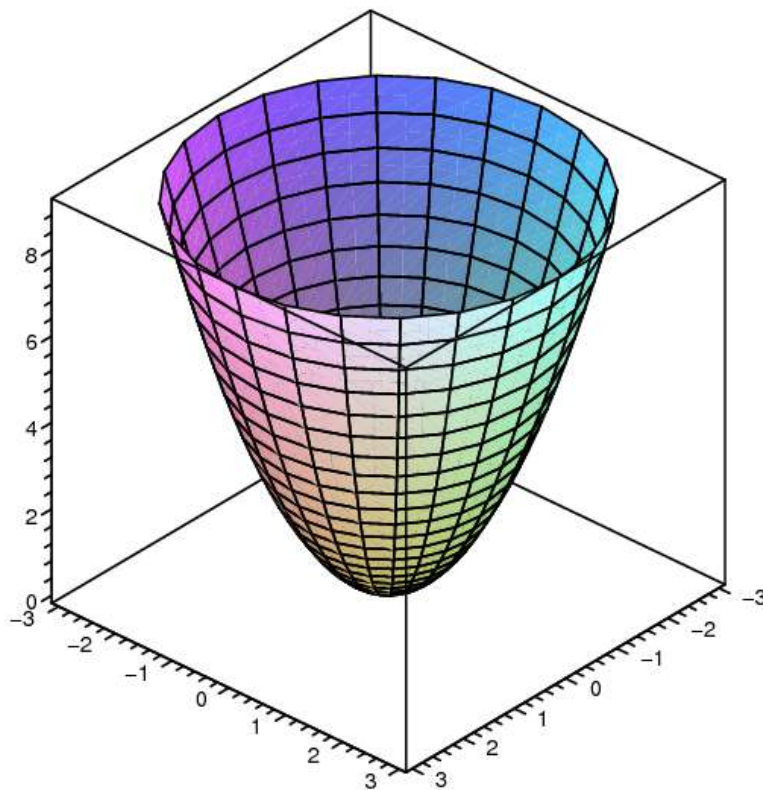
Dealing with functions of several variables

ex: $f(x, y) = x^2 + y^2$

→ describe a surface in 3D

* Domain in $\Rightarrow$ 2D

* output in $\Rightarrow$ 1D "height"



* For MultiVariable Calculus

Partial Derivatives

$$f_x(x, y) = \frac{\partial f}{\partial x}$$

$$f_y(x, y) = \frac{\partial f}{\partial y}$$

Multiple Integral

$$\iint_R f(x, y) dA$$

"Double Integral"

$$\iiint_V f(x, y, z) dV$$

"Triple Integral"

note

* Gradient Vector:

→ is a vector of partial derivatives.

$$\nabla f(x, y, z) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

⇒ Points in the Direction of steepest increase
"الاتجاه الذي يزداد فيه"

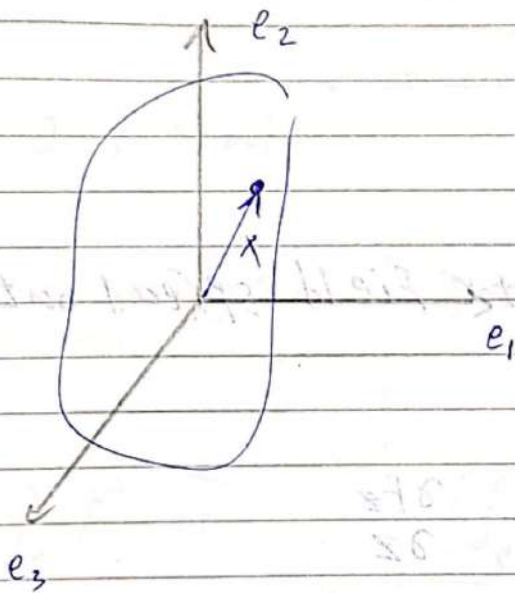


* Vector Calculus :

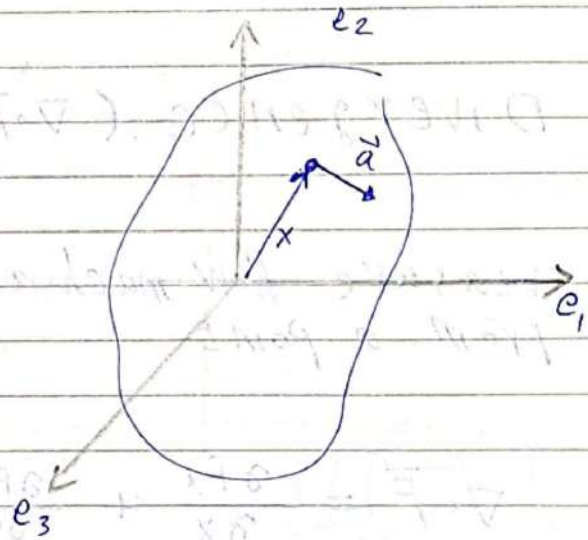
→ study How quantities that have: magnitude Direction change and accumulate in space.

Note :

(Scalar function)



(Vector field)



⇒ give a single value at each point

⇒ give a vector at each point



* Core operations :

① Gradient (∇f) :

→ produce a vector pointing in the Direction of steepest ascent.

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

② Divergence ($\nabla \cdot \vec{F}$) :

→ measure How much a vector Field spread out from a point.

$$\nabla \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

$$\vec{\nabla} \cdot \vec{F} > 0$$

⇒ flowing out

"source"

$$\vec{\nabla} \cdot \vec{F} < 0$$

⇒ flowing in

"sink"



③ curl ($\vec{\nabla} \times \vec{F}$) :

↳ measures a rotation of a vector field around a point.

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

④ Integrals :

line Integral

Surface integral

Volume Integral

$$\int_C \vec{F} \cdot d\vec{r}$$

$$\iint_S \vec{F} \cdot d\vec{s}$$

$$\iiint_V \vec{F} \cdot d\vec{V}$$

Integration
along a curve

Integration
over a surface

Integration
over a Volume



* Tensor calculus :

→ is a generalization of Vector calculus to higher-order objects (Tensors)

0th-order Tensor ← scalar
 1st-order Tensor ← vector
 2nd-order " ← matrix
 Higher-order Tensor ←

▶ dealing with functions where o/p is not just a vector but a multi-dimensional Array of values ..

Scalar Vector Matrix Tensor

1

$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

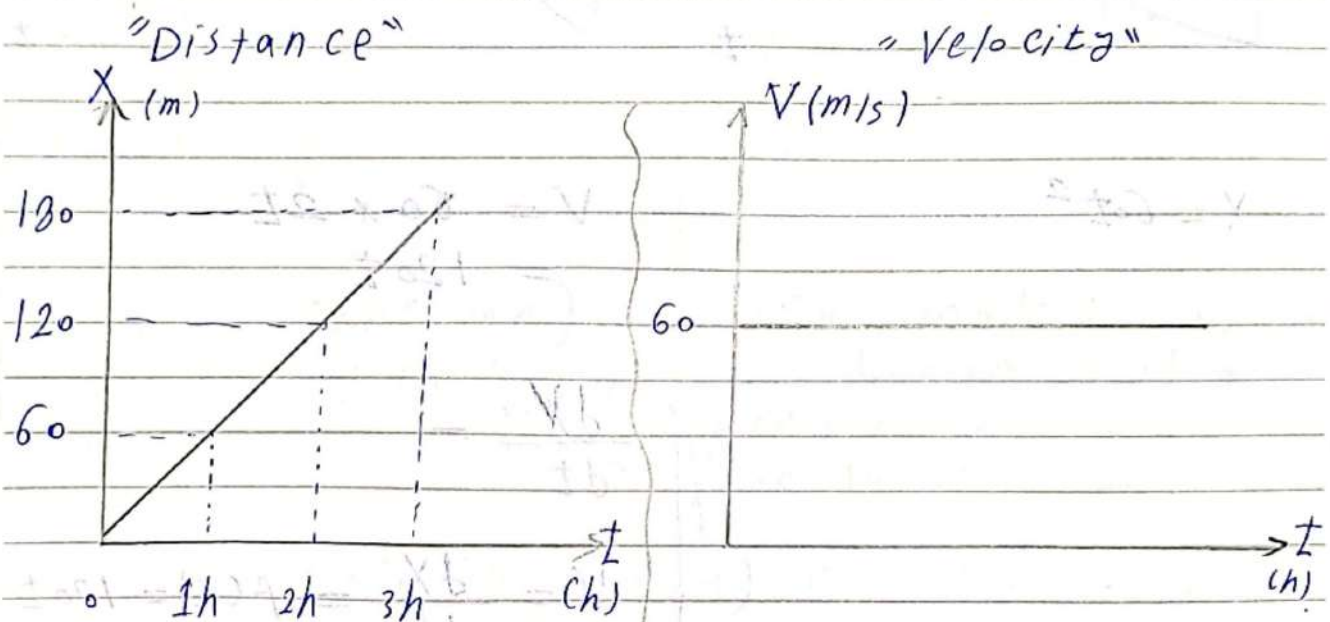
$\begin{bmatrix} 1 & 2 \\ 1 & 7 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 5 & 4 \end{bmatrix}$



* Rate of change :

→ tells us How one quantity changes in relation to another.

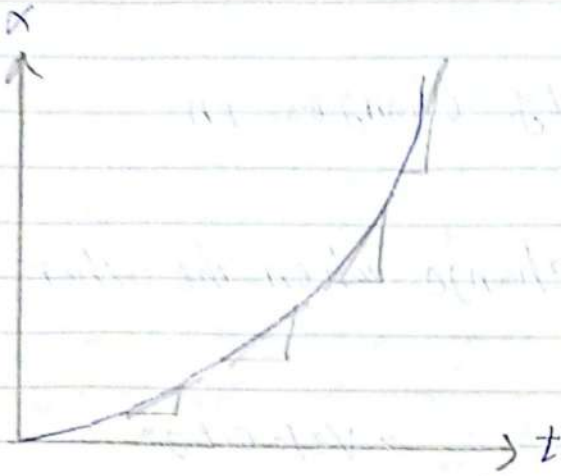
"How fast the variable change when the other one changes".



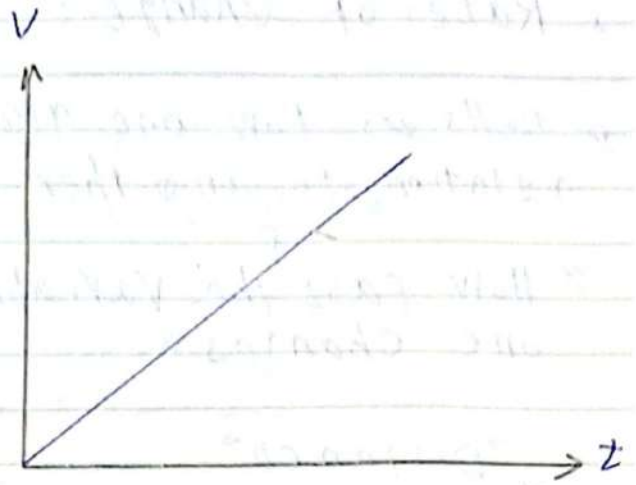
$$X = f(t) = 60t \quad \rightarrow \quad V = \frac{dX}{dt} = 60$$

Rate of change





$$X = 60t^2$$



$$V = 80 \times 2t \\ = 120t$$

$$\frac{dX}{dt} =$$

$$V = \frac{dX}{dt} = f(t) = 120t$$

Instantaneous
Rate of Change

⇒ the rate of change
at one exact point.



* why do we need calculus in ML & DL:

① optimization (training models)

X_1	X_2	...	X_m	$y_{o/p}$

* كل model يكون له (Loss function) وظيفتها حساب الفرق بين prediction و true value

الهدف : تقليل (loss function)

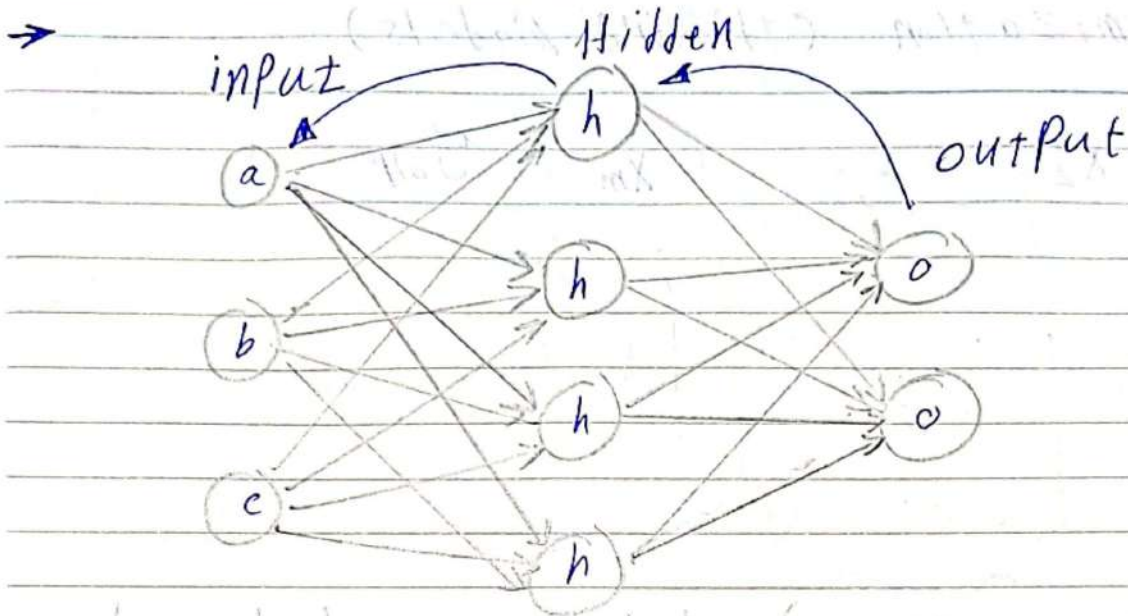
و هذه سيتم بمعرفة اتجاه التغير في ال (model parameters)

* using Gradient Descent

$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$$



② Back Propagation:



ال (Neural Networks) ينتقل عن طريق
 ال Back Propagation لمعرفة ال error في
 كل (layer).

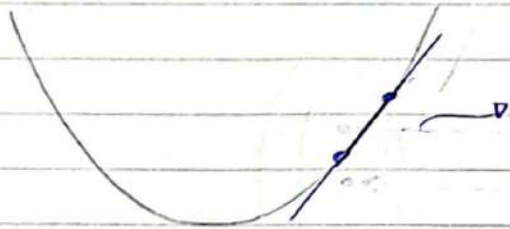
لذلك يستخدم chain Rule

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial w}$$



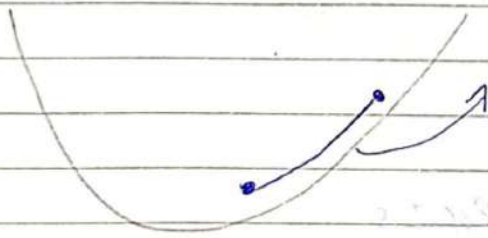
* Note *

① Linear step :



linear step
1st order Approximation
1st derivative.

② Non-linear step :



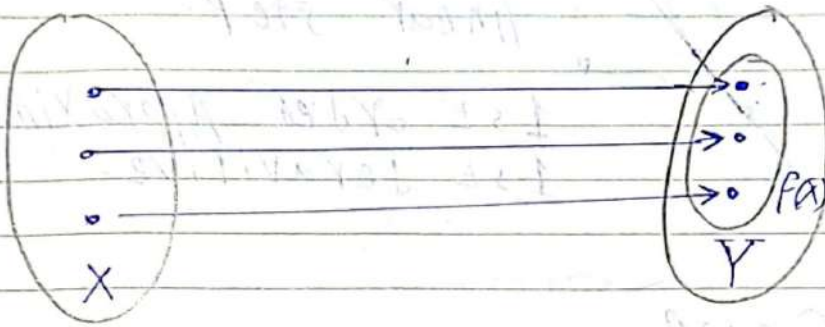
curve step
2nd order Approximation
2nd derivative

خطوة أسرع



* Functions :

→ Relates each element of a set with exactly one element of another set.



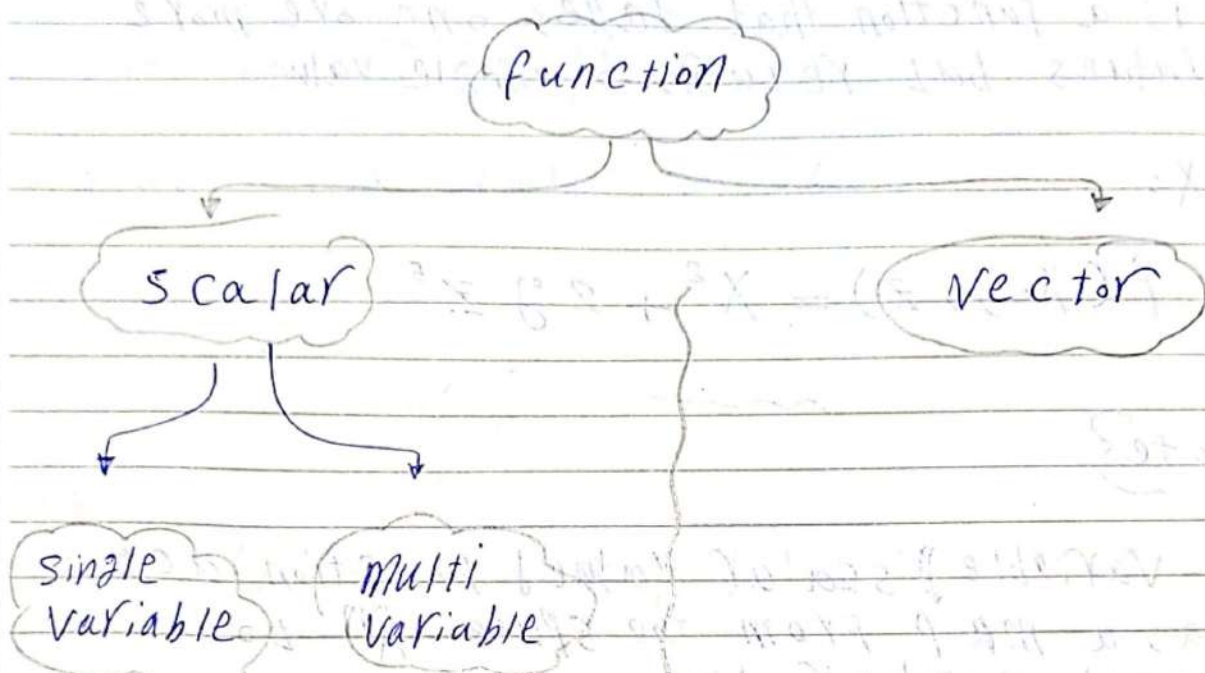
X = Domain

Y = Co-domain

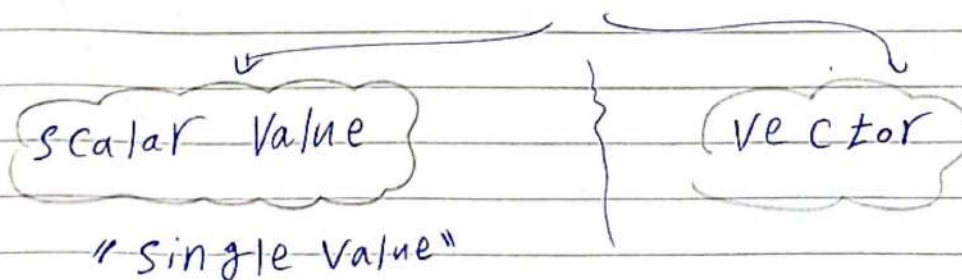
Range = set of outputs.



* Types of Functions :



the output of
the function is:



1. Scalar (Real) Valued Function

→ is a function that takes one or more values but returns → single value

ex:

$$f(x, y, z) = x^2 + 2yz^5$$

Note

((n-variable)) scalar valued function act as a map from the space R^n to real number line.

$$f: R^n \rightarrow R$$



(I) Single-valued function (singular)

input : one variable

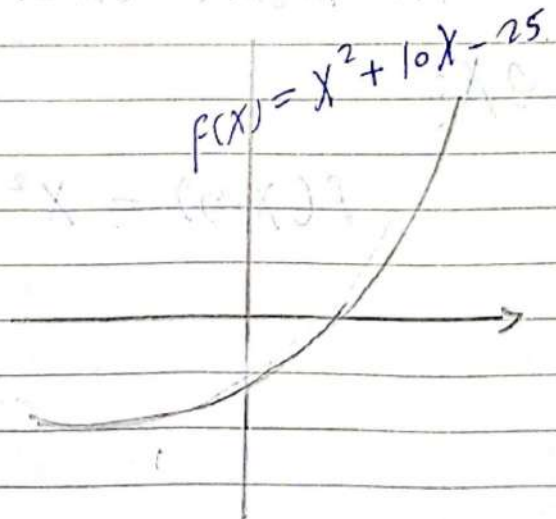
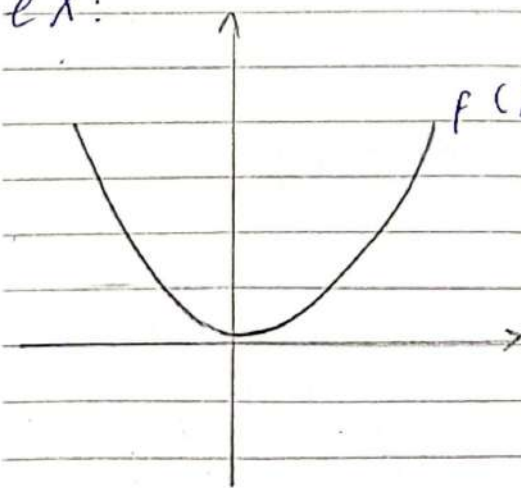
output : one scalar (number)

$$f(x) : \mathbb{R} \rightarrow \mathbb{R}$$

ex:

$$f(x) = x^2 + 3x + 2$$

ex:



II Multi-variable function :

input : two or more variable

output : one scalar (number)

Form :

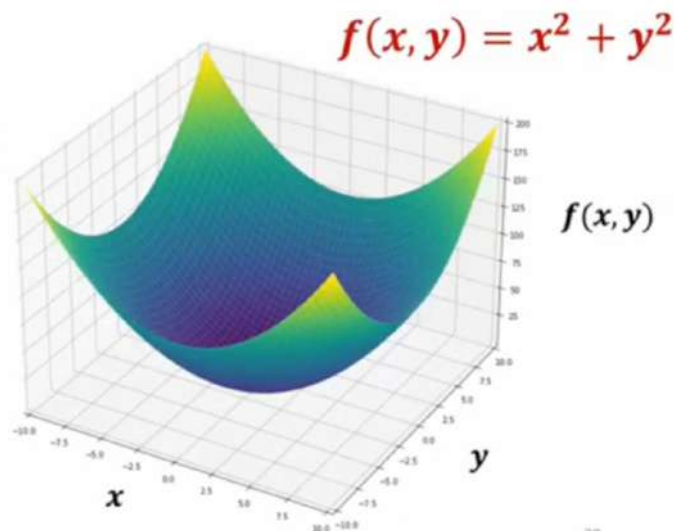
$$f(x_1, x_2, x_3, \dots, x_n) : \mathbb{R}^n \rightarrow \mathbb{R}$$

Note :

it define surfaces or hypersurfaces in higher dimensions.

ex :

$$f(x, y) = x^2 + y^2$$

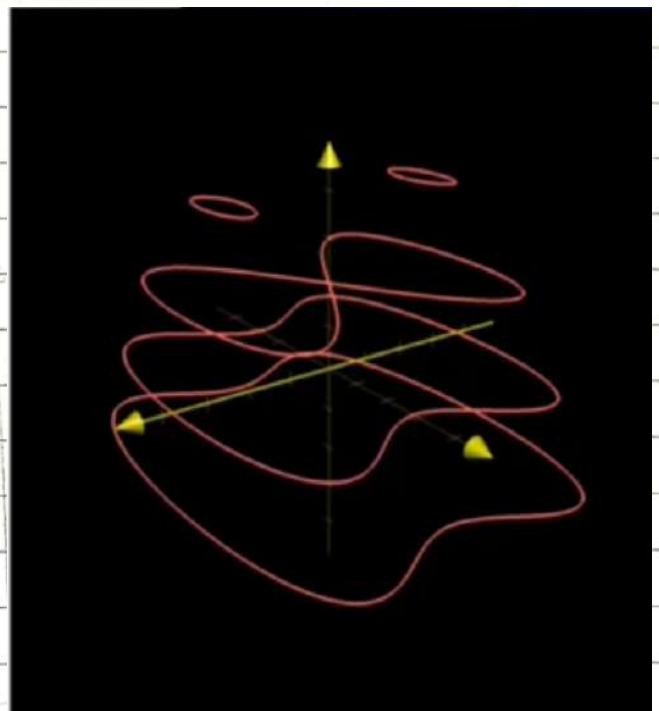
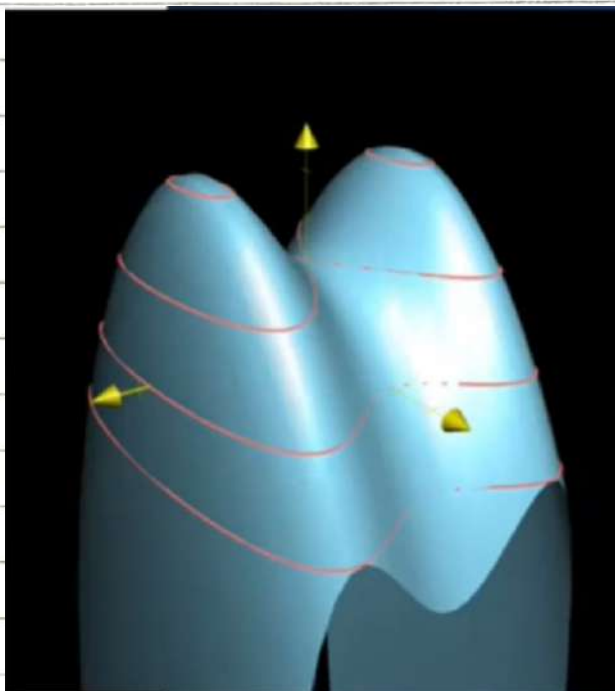


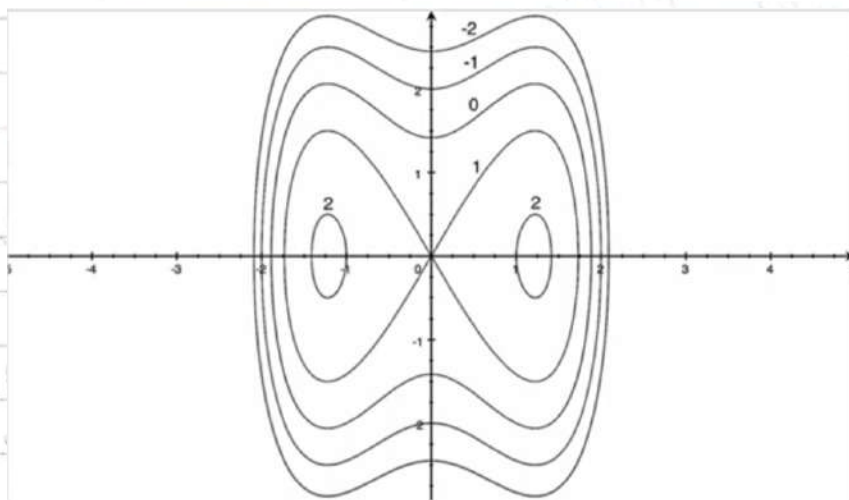
* Contour Plot:-

↳ level curve plot

→ is a way to Visualize a multiVariable scalar function $f(x, y)$ in 2D.

x Sometimes plotting in 3D is hard.
So instead, we Flatten into 2D
using Contours.





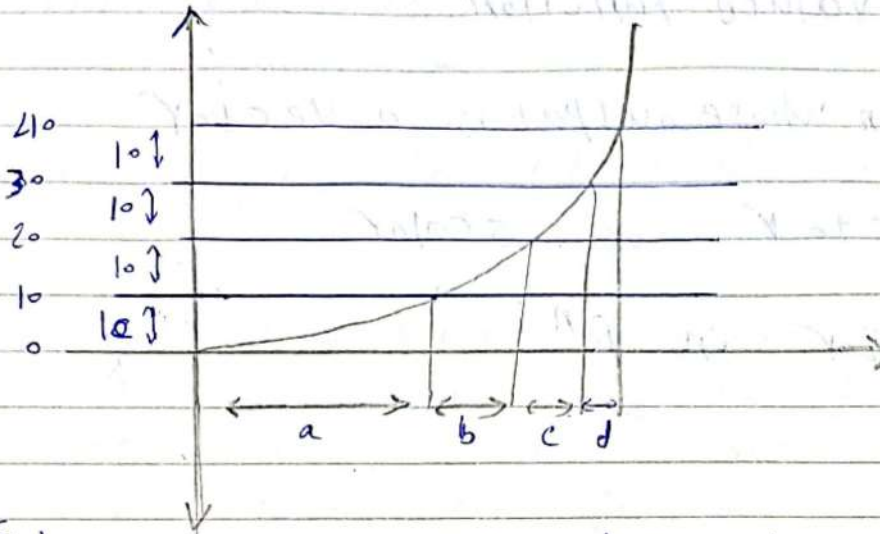
Note

each contour line $\Rightarrow$ is the set of all points (x, y) where function has the same value

$$f(x, y) = \text{const.}$$

x each contour line $\Rightarrow$ different value of const.



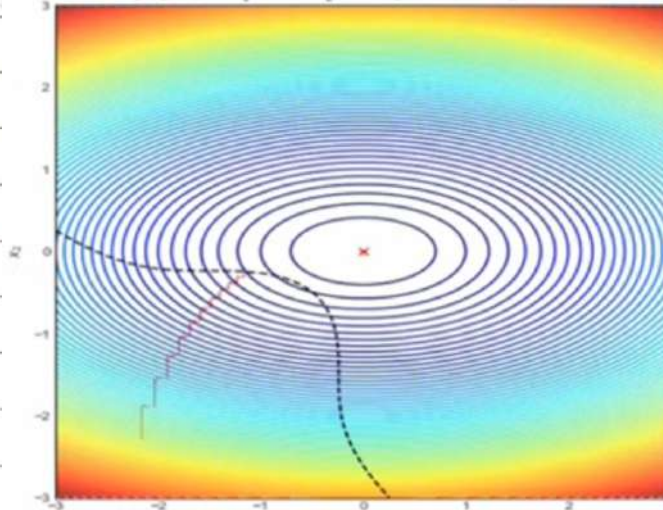


* لاحظ انه كلما ساءت المسافة أقصر معدل ما ساء العيل أكبر
يعني التغير هياكون أسرع

* معدل التغير في الفترة (d) أكبر من معدل التغير عند (a)

* تطبيق في ال ML ؟

في ال optimization باحتاج أعرف اي الاحداثيات
الى عندها المعدل التغير أسرع (المسافات أقصر)
بحيث أوجد ال (optimum value) parameters أسرع
» optimization



[2] Vector Valued function:

→ is a function whose output is a vector.

input: Vector or scalar

output: Vector in $\mathbb{R}^n$

form:

$$\mathbf{r}(t) = \langle f_1(t), f_2(t), \dots, f_n(t) \rangle$$

ex.

① 2D curve

$$\mathbf{r}(t) = \langle \cos(t), \sin(t) \rangle$$

② 3D curve:

$$\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$$

③ physics: (position vector)

$$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$$

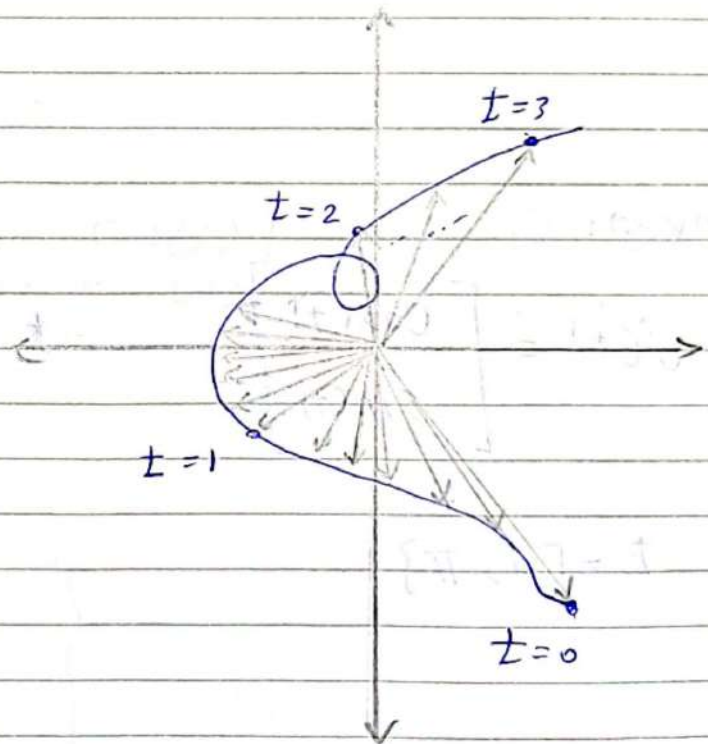


* Parametric function :

→ From (time) to (Location)

→ to model How a particle moves through space over time, you could use a function which takes in one number (t) and outputs the coordinates of the particle.

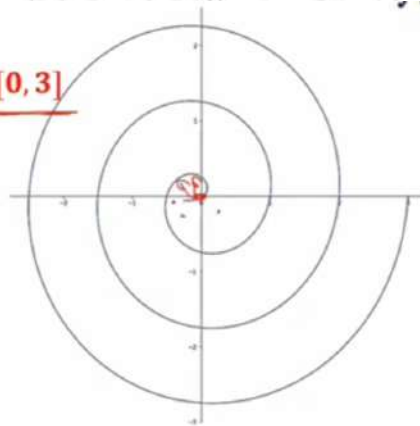
$$f(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$



exo:

$$f(t) = \begin{bmatrix} t \cdot \cos(2\pi t) \\ t \cdot \sin(2\pi t) \end{bmatrix} \quad t = [0, 3]$$

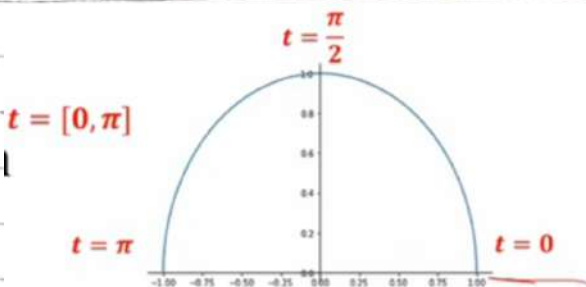
Parametric function

 $t = [0, 3]$ 

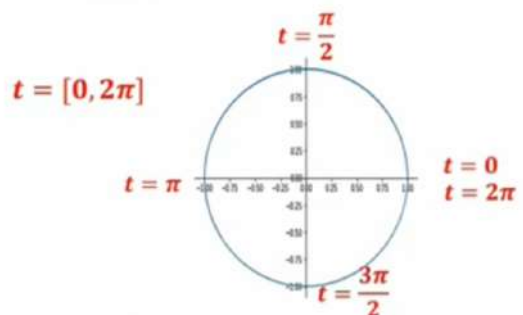
exo:

$$f(t) = \begin{bmatrix} \cos(t) \\ \sin(t) \end{bmatrix} \quad t = [0, \pi]$$

$$t = [0, \pi]$$



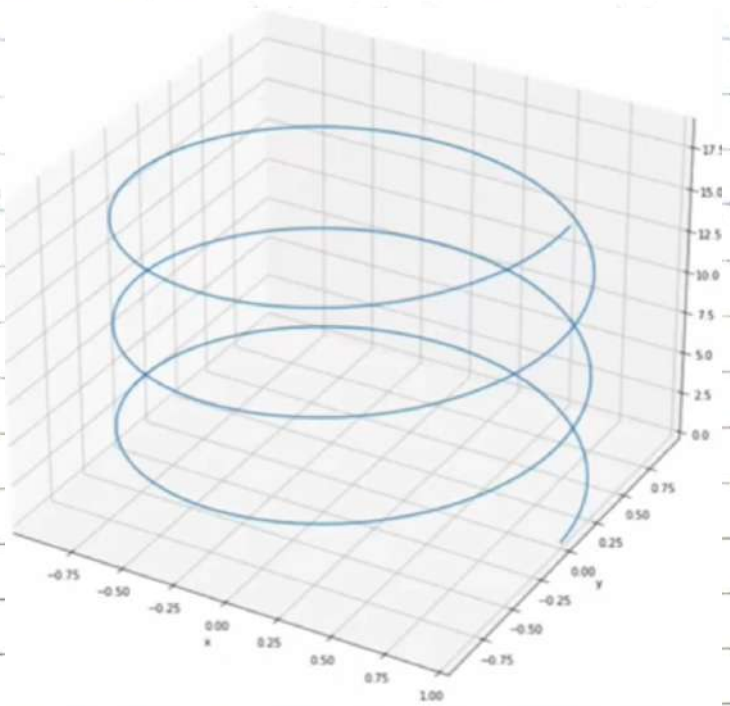
$$t = [0, 2\pi]$$



ex ①

$$f(t) = \begin{bmatrix} \cos(t) \\ \sin(t) \\ t \end{bmatrix}$$

$$t = [0, 6\pi]$$

ex ②

$$f(t,s) = \begin{bmatrix} 3\cos(t) + \cos(t)\cos(s) \\ 3\sin(t) + \sin(t)\cos(s) \\ \sin(s) \end{bmatrix}$$

$$t = [0, \pi]$$

$$s = [0, 2\pi]$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = (0, 1, 1)$$

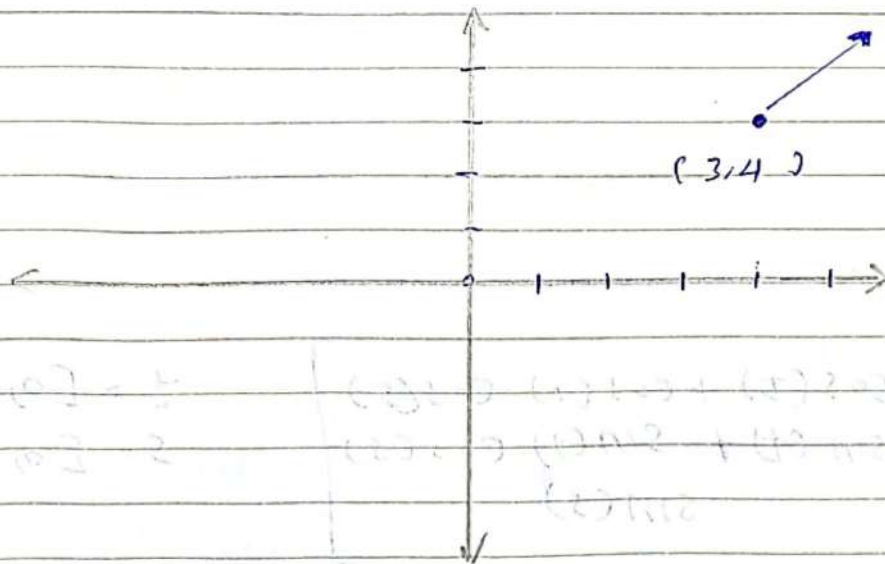
$$(910) \text{ mib} = (910) \text{ mib}$$



* Vector Field:

→ is a function that assigns a vector to every point.

magnitude — Direction



$$f(x, y) = \begin{bmatrix} x \\ y \end{bmatrix}$$

Note

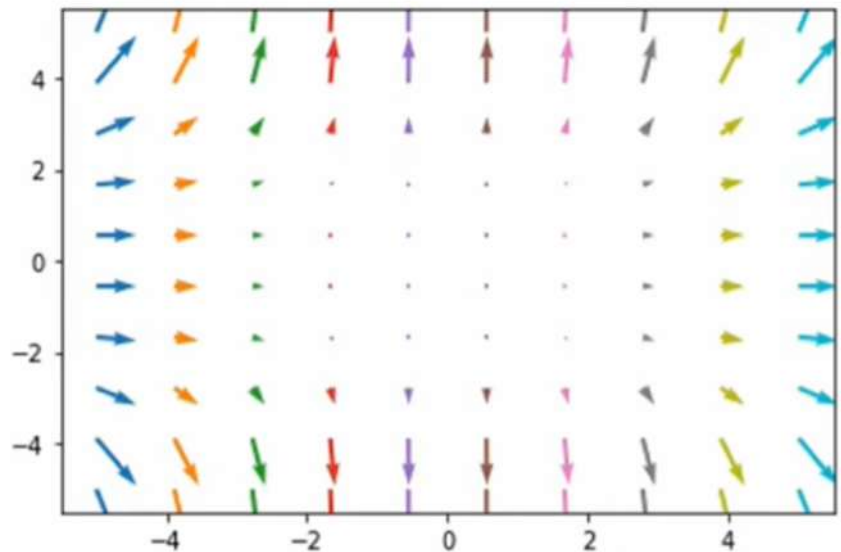
$$\dim(i/p) = \dim(o/p)$$

لذلك نقدر نترسب الـ o/p على الـ i/p



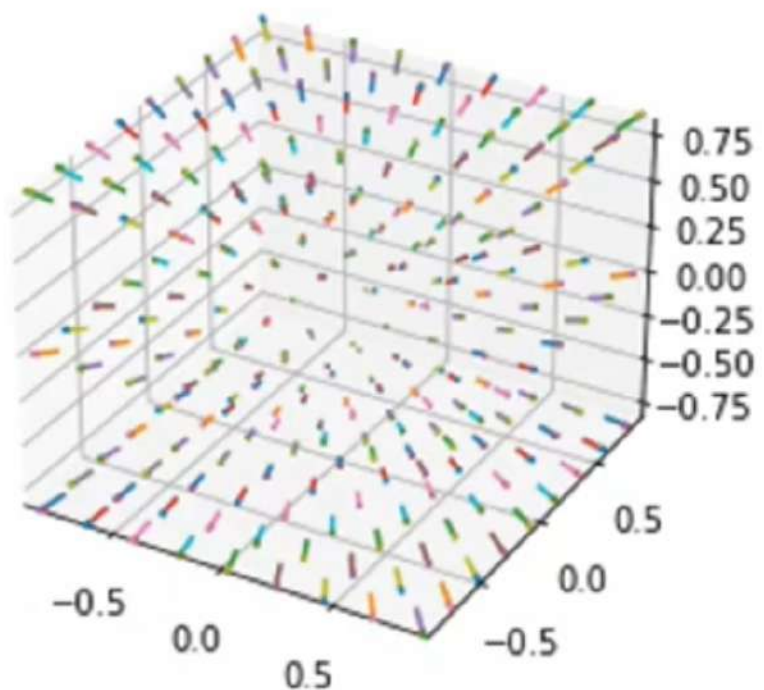
ex①:

$$f(x, y) = \begin{bmatrix} 2x^2 \\ y^2 \end{bmatrix}$$



ex②:

$$f(x, y, z) = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$



* Gradient Vector Field:

→ Helps us to understand the link between Function derivative and the location of min/max. which is crucial in solving optimization problems.

① 2D

$$f(x, y)$$



$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$$

② 3D

$$f(x, y, z)$$

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

ex: $f(x, y) = x^2 + y^2$

$$\nabla f(x, y) = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$



* properties :

① Direction :

→ points in the Direction of the steepest increase of f .

× دايما ال Gradient يشير في اتجاه أسرع زيادة في قيمة الدالة f .

② Magnitude :

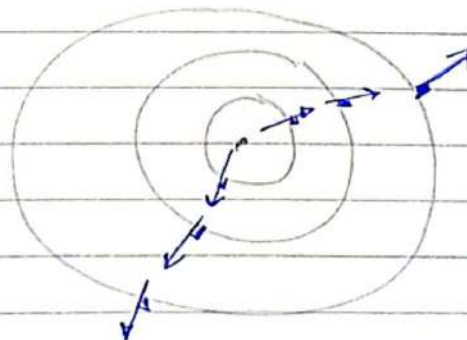
$|\nabla f| \Rightarrow$ the rate of increase in that direction.

Note :

Larger magnitude = faster change

③ orthogonality :

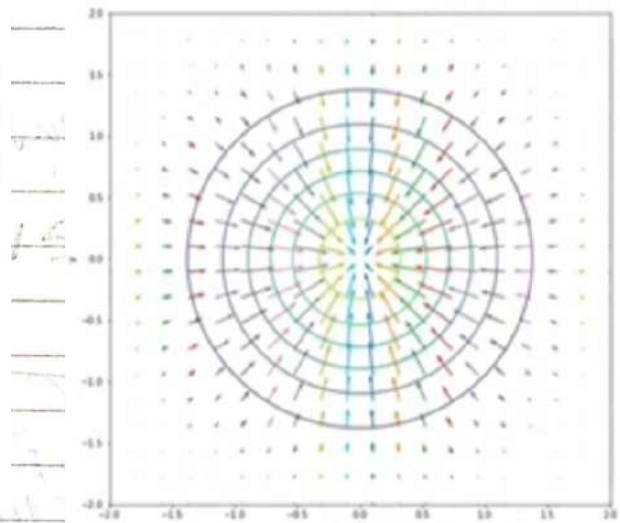
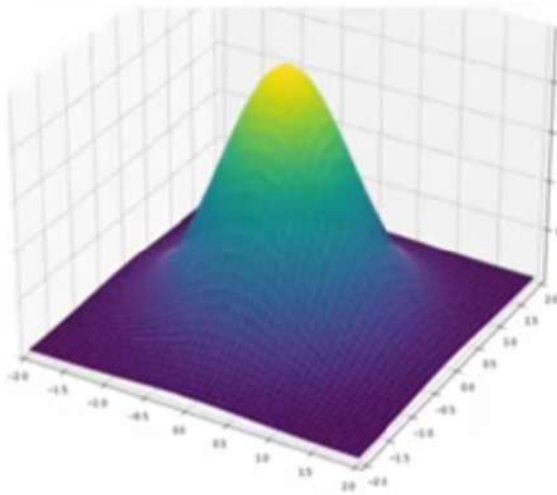
→ the gradient vector is always perpendicular to the level curves of f . (normal)
"contour lines"



ex:

$$f(x, y) = e^{-(x^2 + y^2)}$$

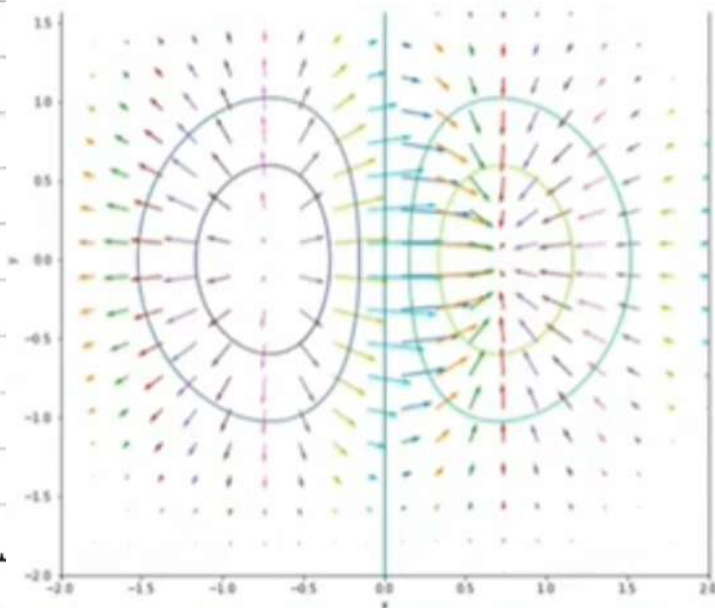
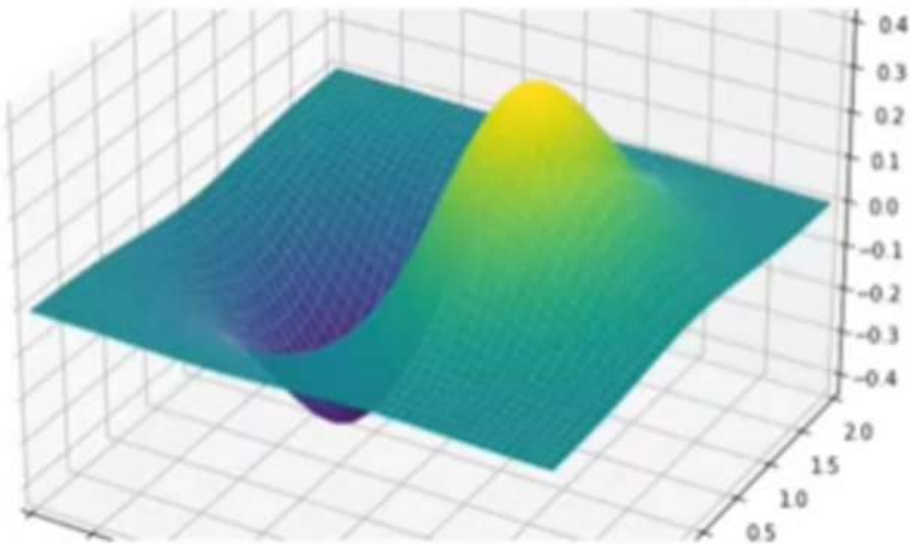
$$\nabla f(x, y) = \begin{bmatrix} -2x e^{-(x^2 + y^2)} \\ -2y e^{-(x^2 + y^2)} \end{bmatrix}$$



ex:

$$f(x, y) = x e^{-(x^2 + y^2)}$$

$$\nabla f(x, y) = \begin{bmatrix} (1 - 2x^2) e^{-(x^2 + y^2)} \\ -2yx e^{-(x^2 + y^2)} \end{bmatrix}$$



* FUNCTION TRANSFORMATION :-

→ is a way to change the position, size or orientation of a function's graph without changing its basic shape

* Main Types :-

① Vertical transformation :-

* Vertical shift :

$$f(x) + c$$

② Vertical stretch / compression :

$$a \cdot f(x)$$

→ $0 < a < 1 \Rightarrow$ compression

→ $a < 0 \Rightarrow$ reflection + $\left\{ \begin{array}{l} \text{comp.} \\ \text{stretch} \end{array} \right.$



② Horizontal transformation:

a) Horizontal shift:

$$f(x - c)$$

b) Horizontal stretch/compression:

$$f(bx)$$

$\rightarrow 0 < b < 1 \Rightarrow$ stretches horizontally

③ Reflections:

• Across the x-axis

$$\rightarrow -f(x)$$

$$\rightarrow f(-x)$$



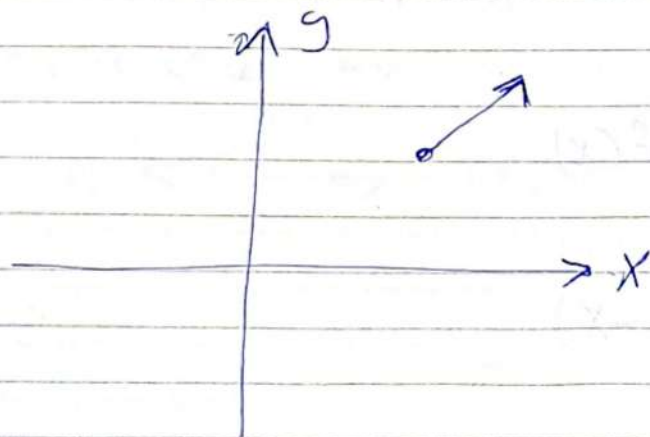
Note

* Linear transformation is a special case of function transformation
"vector valued function"

$$f(x, y) = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

$$= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

* أقدّر أروعهم وال Vector field



ex 1 :

Transformations

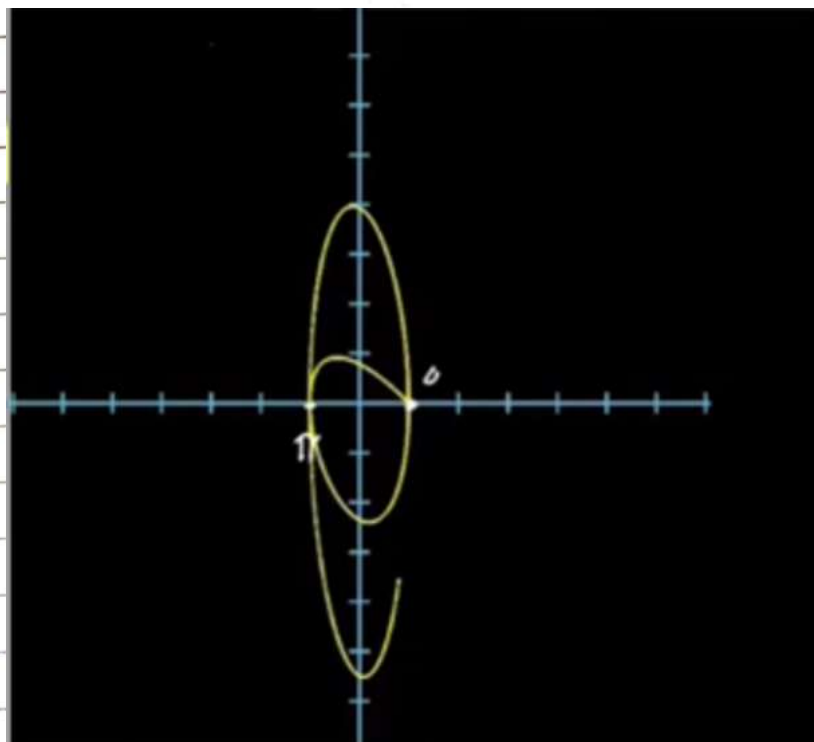
$$f(x) = \begin{bmatrix} \cos(x) \\ x \sin(x) \end{bmatrix}$$

$$f(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$



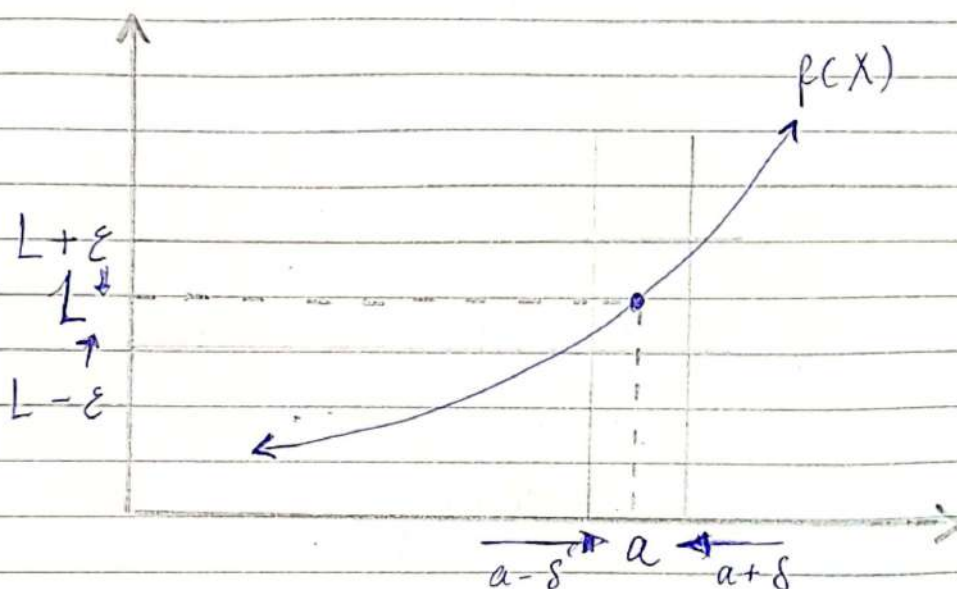
$$f(\pi) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

ex 2 :



* limits:

the limit of the function describes the value that the function approaches as the input gets closer and closer to a certain point.



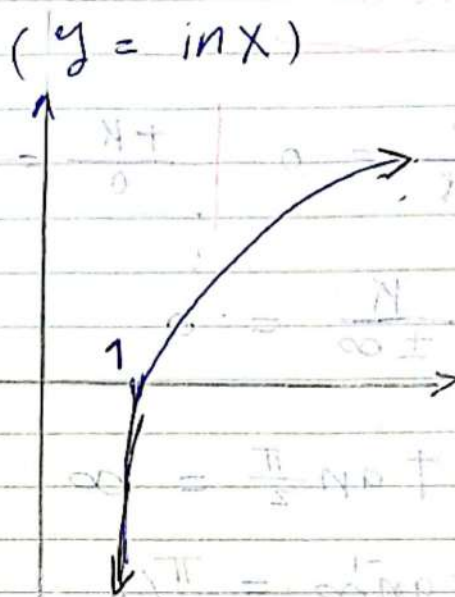
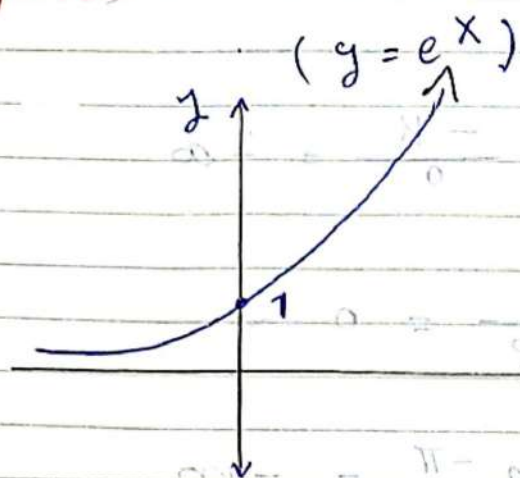
$$\lim_{x \rightarrow a} f(x) = L$$

δ : How close x has to be to (a) in order to $f(x)$ be within $(L \pm \epsilon)$

ϵ : How close we want $f(x)$ to be to L
 "error" or "Tolerance"



(note)



$$e^0 = 1$$

$$e^\infty = \infty$$

$$e^{-\infty} = 0$$

$$\ln e^x = x$$

$$e^{\ln x} = x$$

$$\ln 1 = 0$$

$$\ln \infty = \infty$$

$$\ln 0 = -\infty$$

* properties of $(\ln(x))$:

$$① \quad \ln(x)^n = n \ln(x)$$

$$② \quad \ln(xy) = \ln(x) + \ln(y)$$

$$③ \quad \ln\left(\frac{x}{y}\right) = \ln(x) - \ln(y)$$



(Note)

$$\frac{0}{K} = 0$$

$$\frac{+K}{0} = +\infty$$

$$\frac{-K}{0} = -\infty$$

$$\frac{K}{\pm\infty} = 0$$

$$\frac{0}{\pm\infty} = 0$$

$$\tan \frac{\pi}{2} = \infty, \quad \tan \frac{-\pi}{2} = -\infty$$

$$\tan^{-1} \infty = \pi/2, \quad \tan^{-1} -\infty = -\pi/2$$

$$\tan(0) = 0, \quad \sin(0) = 0, \quad \cos(0) = 1$$

(Note)

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\sin ax}{bx} = \frac{a}{b}$$

$$\lim_{x \rightarrow 0} \frac{\tan ax}{bx} = \frac{a}{b}$$

$$\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx} = \frac{a}{b}$$

$$\lim_{x \rightarrow 0} \frac{ax}{\tan bx} = \frac{a}{b}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$



$$\lim_{x \rightarrow a} f(x)$$

① التحويلات من العبادات

①

ثابتة

+ve -ve 0

$\Rightarrow$ limit exists

النهاية موجودة

(II)

$(\pm \infty)$

limit :

doesn't exist

النهاية غير موجودة

المنبأ لانه فاصلة

(III)

نسبة غير معينة

$\Rightarrow \frac{0}{0}, \frac{\infty}{\infty}$

$0 \times \infty$

$\infty - \infty$

$\infty^0, 0^0, 1^\infty$

$\Rightarrow$ (l'Hopital rule)

ex:

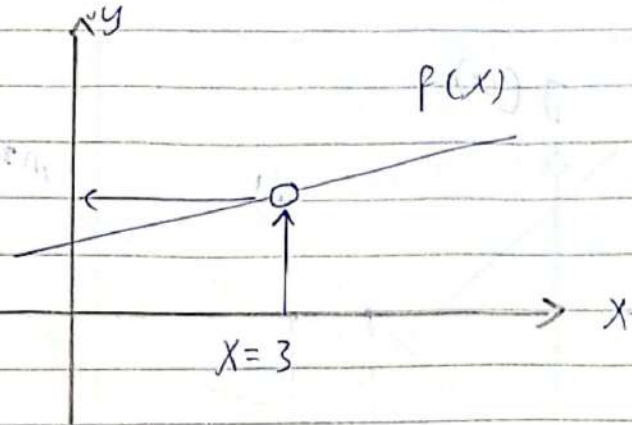
$$\textcircled{1} \lim_{x \rightarrow 2} \frac{x-2}{x+3} = \frac{2-2}{2+3} = 0 \rightarrow \text{exist}$$

$$\textcircled{2} \lim_{x \rightarrow 1} \frac{x^2-4}{x-1} = \frac{1-4}{1-1} = \frac{-3}{0} = -\infty \rightarrow \text{doesn't exist}$$

$$\textcircled{3} \lim_{x \rightarrow 2} \frac{x^2-4}{x-2} = \frac{4-4}{2-2} = \frac{0}{0} \Rightarrow \text{uncertain quantity}$$

(l'Hopital)

* ex:



$$y = ax + b$$

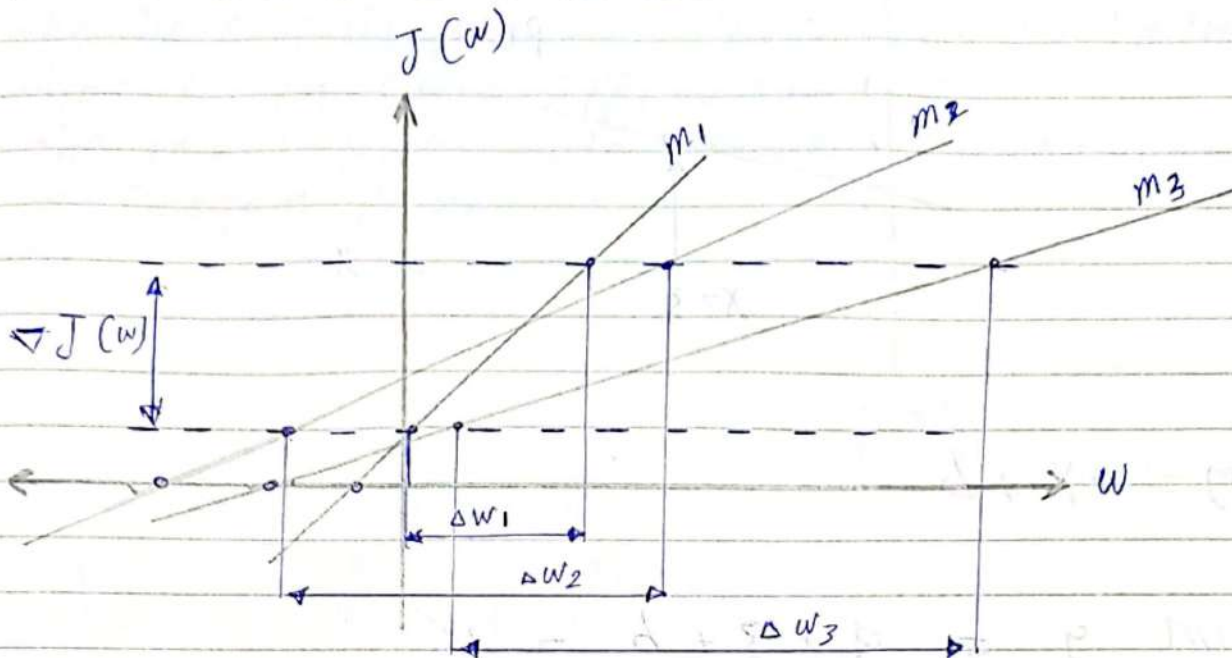
$$\lim_{x \rightarrow 3} y = a \times 3 + b = \checkmark$$

بما أنه y discontinuous عند $x=3$

لم إذاً أنا مخرجت أحسب قيمة الـ y عندها بشكل مباشر لذلك ← بحسب القيمة هنا خلال الـ limit



* Gradient's Intuition:



$$m_1 = \frac{\nabla J(w_1)}{\nabla w_1}, \quad m_2 = \frac{\nabla J(w_2)}{\nabla w_2}, \quad m_3 = \frac{\nabla J(w_3)}{\nabla w_3}$$

1) We assume $\nabla J(w_1) = \nabla J(w_2) = \nabla J(w_3)$

2) and $m_1 > m_2 > m_3$

then:

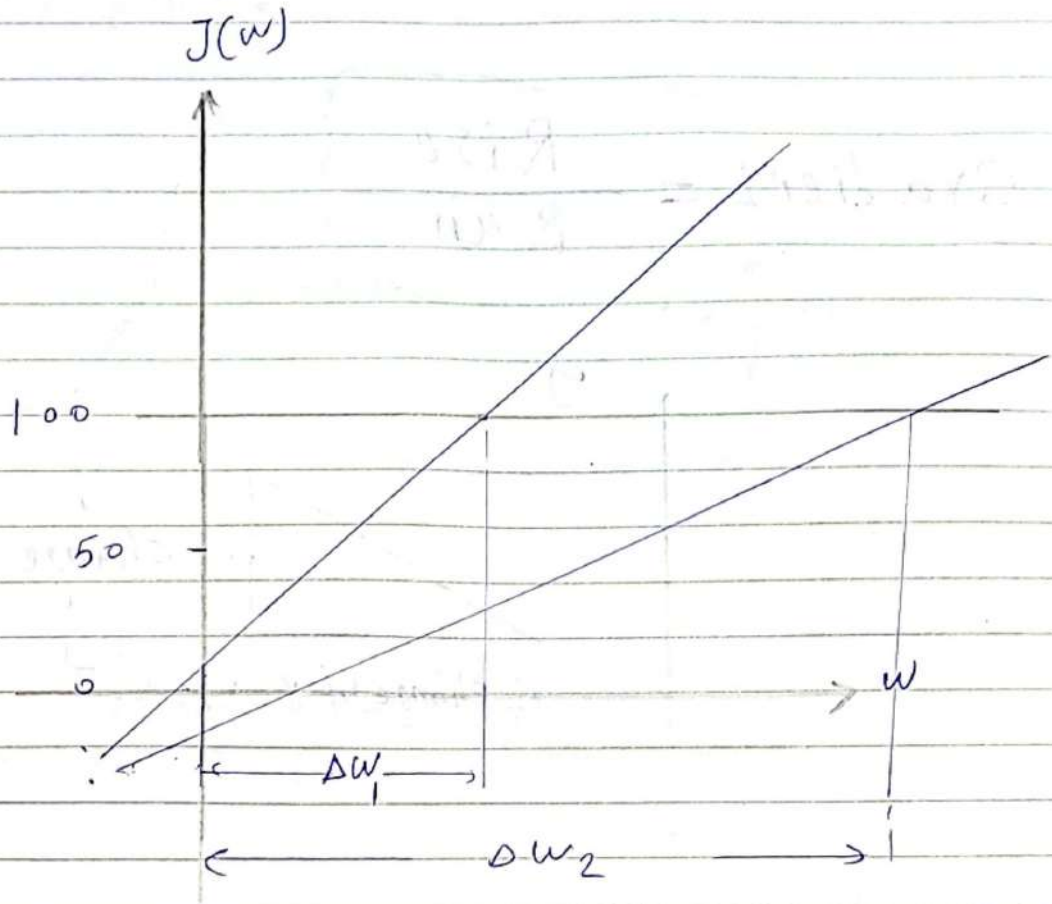
$$\nabla w_1 < \nabla w_2 < \nabla w_3$$

يعني مع تزايد الـ $\nabla J(w)$ كلما كان الـ slope أكبر، كلما كانت معدل التغير أكبر، كلما كانت الـ (sensitivity) أكبر.

أي: تتأثر بأقل تغير في الـ w



ex:



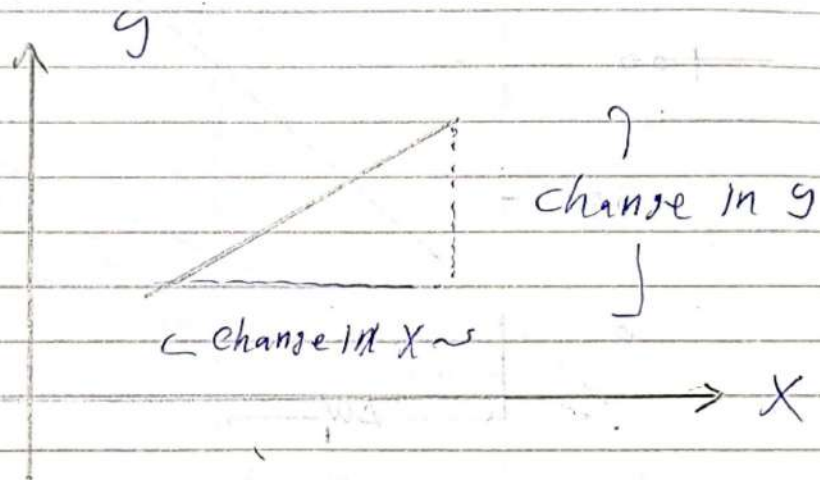
للمحول $J(w) = 100$ إلى

أزهر التغير بمقدار (Δw_1) في توصل أسرع
 لأن slope أكبر

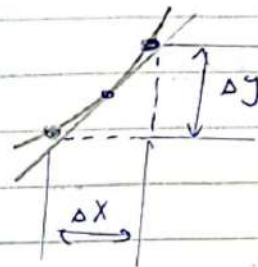
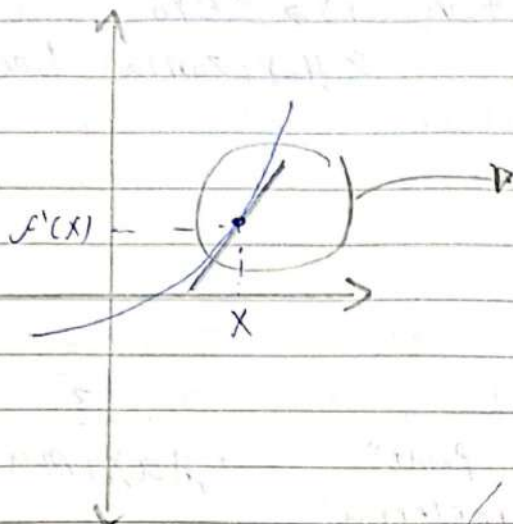
أو
 التغير بمقدار (Δw_2)



$$\text{Gradient} = \frac{\text{Rise}}{\text{Run}}$$



* Derivative :



$$\frac{\Delta y}{\Delta x} = \frac{f(X + \Delta x) - f(X)}{\Delta x}$$

$$f'(X) = \lim_{\Delta x \rightarrow 0} \left(\frac{f(X + \Delta x) - f(X)}{\Delta x} \right)$$

Note

Gradient at X = derivative at X

لا زلنا نقطتين على الـ Curve قريبتين
من النقطة X على أن يكون الـ derivative valid
الـ tangent line $f'(X)$



* Stationary Point:

is a point where derivative is $\Rightarrow$ Zero

"Horizontal line"

* Types :

[1]

"Minimum"

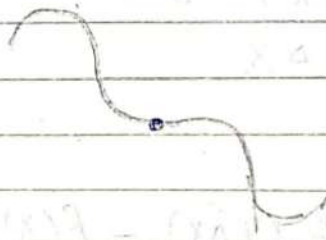


$$f'(x) = 0$$

$$f''(x) > 0$$

[2]

"saddle point"
point of inflection



$$f'(x) = 0$$

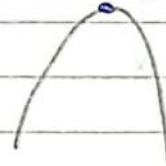
$$f''(x) = 0$$

(or)

= changes
sign ($\pm$)

[3]

"Maximum"



$$f'(x) = 0$$

$$f''(x) < 0$$

ملحوظة :
x اسمها (stationary point) لأنها لم تحركت (لم يتغير موقعها)
هتفعلك يا بنته من هتتحرك



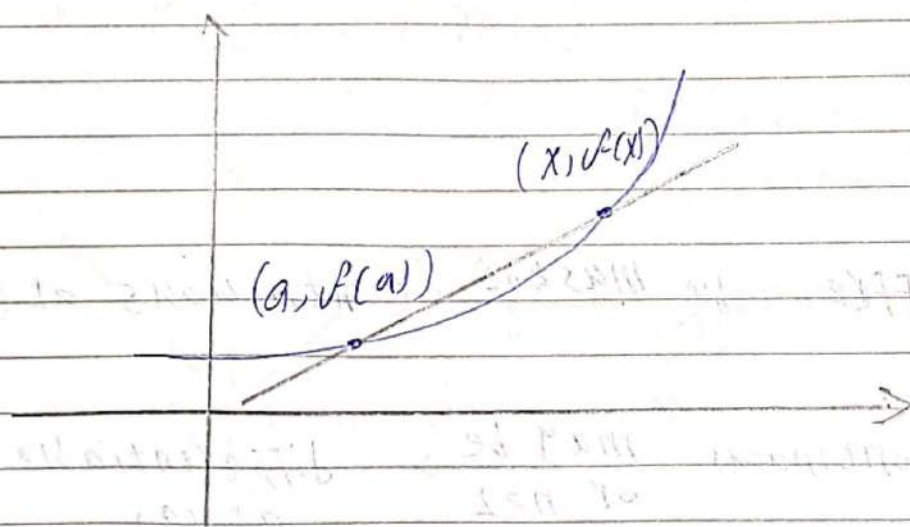
* the secant line:

→ is a line cutting a curve at two points.

→ represent the average of change between two points.

* As the two points get closer

→ the secant becomes the tangent.



$$m_{\text{sec}} = \frac{f(x) - f(a)}{x - a}$$



* mean value theorem * نظرية القيمة المتوسطة

suppose: $f = f(x)$

f is continuous over closed interval $[a, b]$

and

f is differentiable on the interval's interior (a, b) .

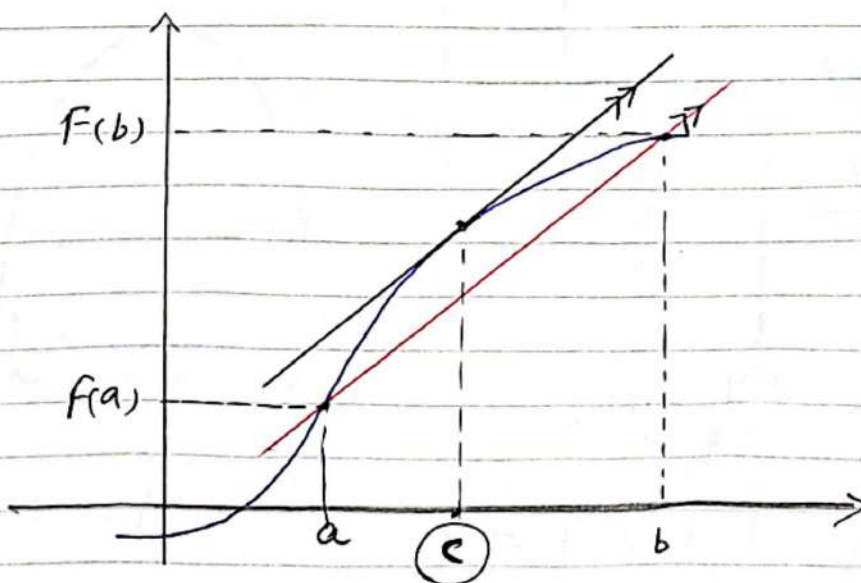
then,

$\Rightarrow$ there is at least one point $c \in (a, b)$

such that $\Rightarrow$

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

في كل المعادلات التي يمر بالنقطة $(a, f(a))$ و $(b, f(b))$ $\Rightarrow$ $f'(c) = \frac{f(b) - f(a)}{b - a}$ $\Rightarrow$ $f'(c) = \frac{f(b) - f(a)}{b - a}$ $\Rightarrow$ $f'(c) = \frac{f(b) - f(a)}{b - a}$



$$m = f'(c)$$



* Differentiability & Continuity:

→ A function $f(x)$ is continuous at point a if and only if:

[1] $f(a)$ is defined

[2] $\lim_{x \rightarrow a} f(x)$ exists

[3] $\lim_{x \rightarrow a} f(x) = f(a)$

Note :

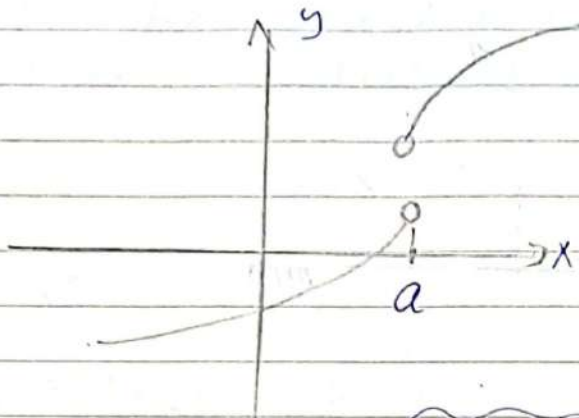
* if $f(a)$ differentiable must be continuous at (a)

* if $f(a)$ continuous may be or not differentiable at (a)



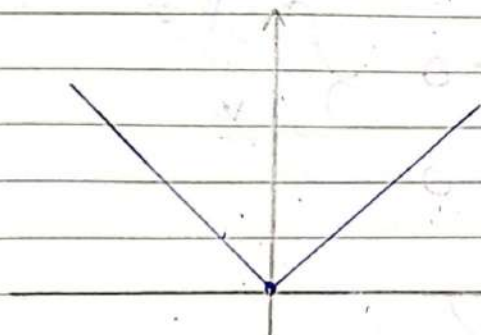
* undefined slope:

① Discontinuity :



$$f'(a) = \text{undefined}$$

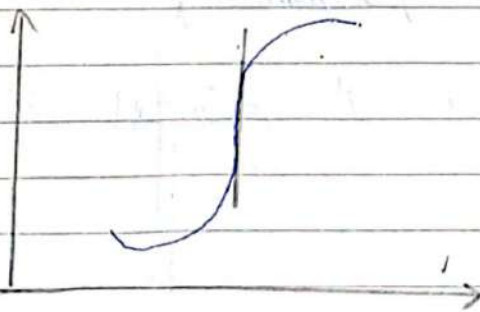
② corner (sharp Turn)



left-hand derivative $\neq$ right-hand derivative

* لأنها نقطة واحدة
 أنا محتاج نقطتين عيّنات
 أخرج أكتب الـ derivative

③ Vertical Tangent line:

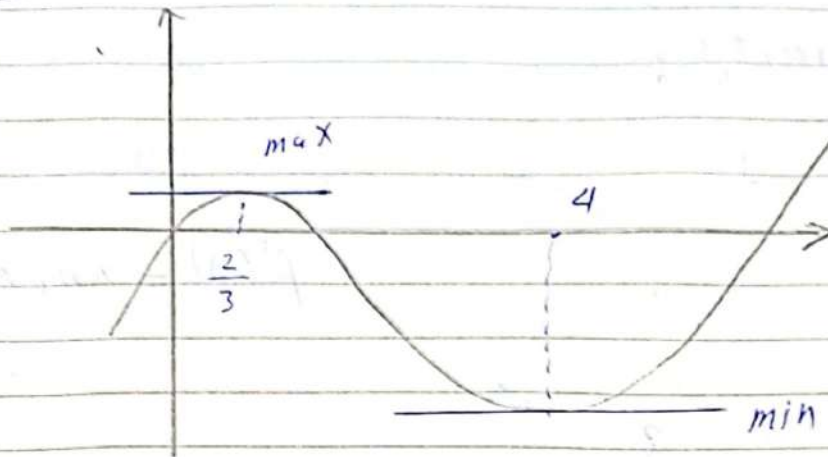


$$\text{slope} = \frac{dy}{dx} = \infty$$

$$\therefore \Delta x = 0$$



* Note :



$$f(x) = x^3 - 7x^2 + 8x + 1$$

$$f'(x) = 3x^2 - 14x + 8 = 0 \quad \left\{ \begin{array}{l} x = \\ x \end{array} \right.$$

$$= (3x - 2)(x - 4) = 0$$

$$x_1 = \frac{2}{3}, \quad x_2 = 4$$

* عنى الـ (high polynomial) متى بقدر ايجادها Analytical

لازم المتقزم الـ (Approximation methods)



GENERAL DERIVATIVE RULES

Constant Rule	$\frac{d}{dx}[c] = 0$
Constant Multiple Rule	$\frac{d}{dx}[cf(x)] = cf'(x)$
Sum Rule	$\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$
Difference Rule	$\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x)$
Product Rule	$\frac{d}{dx}[f(x) \cdot g(x)] = f'(x)g(x) + f(x)g'(x)$
Quotient Rule	$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$
Chain Rule	$\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$

DERIVATIVE RULES FOR PARTICULAR FUNCTIONS

FUNCTION	BASIC RULE	CHAIN RULE
Power	$\frac{d}{dx}[x^n] = nx^{n-1}$	$\frac{d}{dx}[u^n] = nu^{n-1} \cdot u'$
TRIGONOMETRIC FUNCTIONS		
Sine	$\frac{d}{dx}[\sin(x)] = \cos(x)$	$\frac{d}{dx}[\sin(u)] = \cos(u) \cdot u'$
Cosine	$\frac{d}{dx}[\cos(x)] = -\sin(x)$	$\frac{d}{dx}[\cos(u)] = -\sin(u) \cdot u'$
Tangent	$\frac{d}{dx}[\tan(x)] = \sec^2(x)$	$\frac{d}{dx}[\tan(u)] = \sec^2(u) \cdot u'$
Cosecant	$\frac{d}{dx}[\csc(x)] = -\csc(x) \cot(x)$	$\frac{d}{dx}[\csc(u)] = -\csc(u) \cot(u) \cdot u'$
Secant	$\frac{d}{dx}[\sec(x)] = \sec(x) \tan(x)$	$\frac{d}{dx}[\sec(u)] = \sec(u) \tan(u) \cdot u'$
Cotangent	$\frac{d}{dx}[\cot(x)] = -\csc^2(x)$	$\frac{d}{dx}[\cot(u)] = -\csc^2(u) \cdot u'$
INVERSE TRIGONOMETRIC FUNCTIONS		
Arcsine	$\frac{d}{dx}[\sin^{-1}(x)] = \frac{1}{\sqrt{1-x^2}}$	$\frac{d}{dx}[\sin^{-1}(u)] = \frac{1}{\sqrt{1-u^2}} \cdot u'$
Arccosine	$\frac{d}{dx}[\cos^{-1}(x)] = \frac{-1}{\sqrt{1-x^2}}$	$\frac{d}{dx}[\cos^{-1}(u)] = \frac{-1}{\sqrt{1-u^2}} \cdot u'$
Arctangent	$\frac{d}{dx}[\tan^{-1}(x)] = \frac{1}{1+x^2}$	$\frac{d}{dx}[\tan^{-1}(u)] = \frac{1}{1+u^2} \cdot u'$
Arccosecant	$\frac{d}{dx}[\csc^{-1}(x)] = \frac{-1}{ x \sqrt{x^2-1}}$	$\frac{d}{dx}[\csc^{-1}(u)] = \frac{-1}{ u \sqrt{u^2-1}} \cdot u'$
Arcsecant	$\frac{d}{dx}[\sec^{-1}(x)] = \frac{1}{ x \sqrt{x^2-1}}$	$\frac{d}{dx}[\sec^{-1}(u)] = \frac{1}{ u \sqrt{u^2-1}} \cdot u'$
Arccotangent	$\frac{d}{dx}[\cot^{-1}(x)] = \frac{-1}{1+x^2}$	$\frac{d}{dx}[\cot^{-1}(u)] = \frac{-1}{1+u^2} \cdot u'$
EXPONENTIAL FUNCTIONS		
Exponential (base e)	$\frac{d}{dx}[e^x] = e^x$	$\frac{d}{dx}[e^u] = e^u \cdot u'$
Exponential (base a)	$\frac{d}{dx}[a^x] = a^x \ln(a)$	$\frac{d}{dx}[a^u] = a^u \ln(a) \cdot u'$
LOGARITHMIC FUNCTIONS		
Natural Logarithm	$\frac{d}{dx}[\ln(x)] = \frac{1}{x}$	$\frac{d}{dx}[\ln(u)] = \frac{1}{u} \cdot u' \text{ or } \frac{u'}{u}$
Logarithm (base a)	$\frac{d}{dx}[\log_a(x)] = \frac{1}{x \cdot \ln(a)}$	$\frac{d}{dx}[\log_a(u)] = \frac{1}{u \cdot \ln(a)} \cdot u'$



Chain Rule :

→ states that the derivative of a composite function is a derivative of the outer function evaluated at the inner function times the derivative of inner function.

(I)

single-variable

(II)

multi-variable

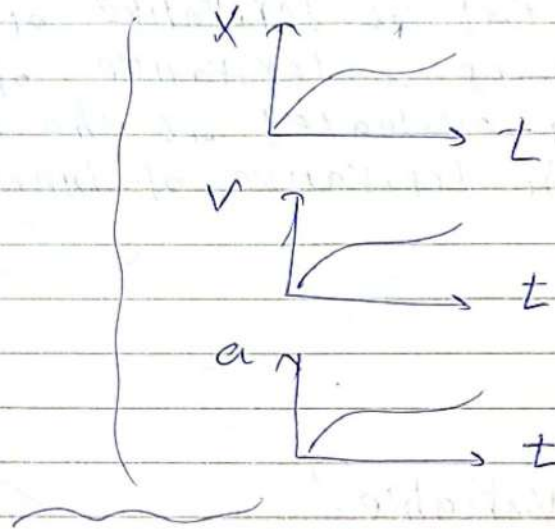


□ single-variable Chain Rule ::

$$X = f(t)$$

$$V = \frac{dX}{dt}$$

$$a = \frac{dV}{dt}$$



$$a \quad \text{---} \quad 1$$

$$\boxed{\frac{da}{dV} = 10}$$

$$V \quad \text{---} \quad 0.4$$

$$\boxed{\frac{dV}{dX} = 0.2}$$

$$X \quad \text{---} \quad 2$$

$$\boxed{\frac{dX}{dt} = 2}$$

$$t \quad \text{---} \quad 1$$

$$\therefore \frac{da}{dt} = \frac{da}{dV} \cdot \frac{dV}{dX} \cdot \frac{dX}{dt}$$



ex :

$$y = f(u) = u^2 + 2$$

$$f(u) = a e^a$$

$$a(u) = \sin(u)$$

$$u(x) = \cancel{5x^4} \quad x^3 + 2x$$

$$f(x) = f(a(u(x)))$$

$$\frac{df}{dx} = \frac{df}{da} \cdot \frac{da}{du} \cdot \frac{du}{dx}$$

$$= f'(a(u(x))) \cdot a'(u(x)) \cdot u'(x)$$



e^x :

$$f(u) = e^u \longrightarrow f'(u) = e^u$$

$$u(x) = 3x^2 + 1 \longrightarrow u'(x) = 6x$$

$$\begin{aligned} \therefore f'(u) &= u' e^u \\ &= 6x e^{(3x^2 + 1)} \end{aligned}$$



× أهمية (chain Rule) :

× مهم في الـ (training) لـ ML & DL

① optimization (Gradient Descent) :

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n$$

$J(\theta)$ = cost function (error func)

× لما نعمل (update) للأوزان في أي model لازم نعرف كيف يتغير الـ loss لما الـ weight يتغير

« مقدار الـ (sensitivity)

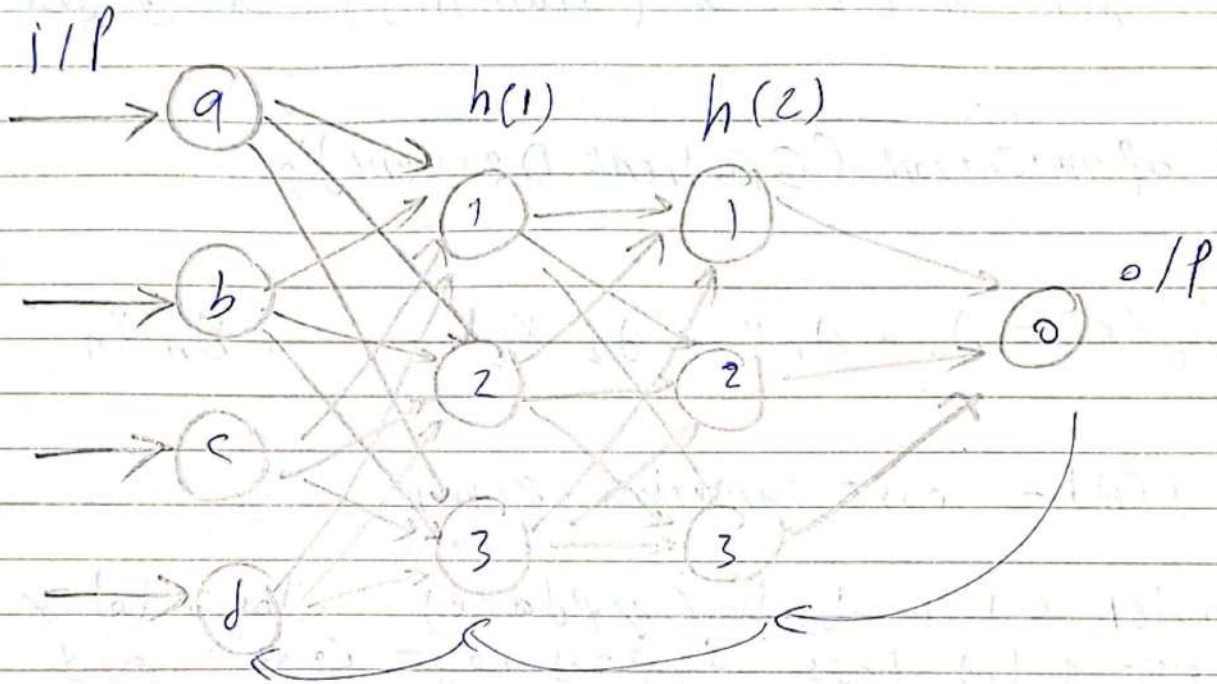
عن طريق الـ (chain Rule)

لأننا الـ loss معتمد على network

network ← $\frac{\text{مقدرة}}{\text{في}}$ weights



② Back propagation :-



بالتسلسل (chain rule) :

$$x \rightarrow h_1(x) \rightarrow h_2(h_1(x)) \rightarrow \dots$$

أقدر أعمل Back propagation
وارجع أعدل في الـ (weight) لتقليل error



* second derivative :-

↳ the derivative of the first derivative

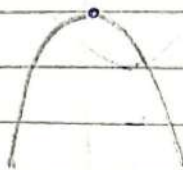
→ tells how the slope itself changes.

Note:

$$f''(x) > 0$$

$$f''(x) < 0$$

$$f''(x) = 0$$



• concave
upward ↑

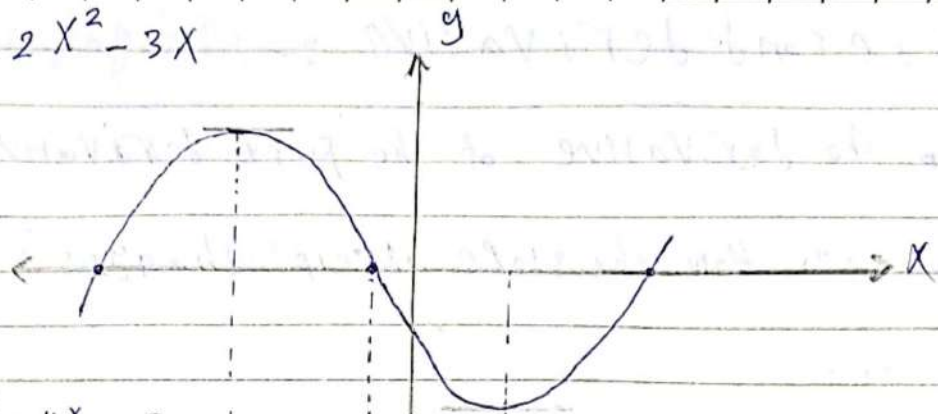
• concave
downward ↓

• Point of
INFlection

⇒ concavity
changes



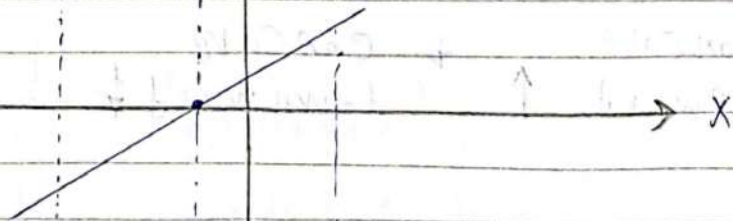
$$f(x) = 5x^3 + 2x^2 - 3x$$



$$f(x) = 15x^2 + 4x - 3$$



$$f'(x) = 30x + 4$$



* Partial Derivatives *

$$Z = f(x, y)$$

$$\rightarrow \left(\frac{\partial f}{\partial x} \right) \Rightarrow$$

بأنه إذا أخذنا المشتقة الجزئية لـ f بالنسبة لـ x مع اعتبار أن y ثابتة.

$$\rightarrow \left(\frac{\partial f}{\partial y} \right) \Rightarrow$$

بأنه إذا أخذنا المشتقة الجزئية لـ f بالنسبة لـ y مع اعتبار أن x ثابتة.

1- First derivative :

$$\Rightarrow \boxed{\frac{\partial f}{\partial x} = f_x}$$

$$\Rightarrow \boxed{\frac{\partial f}{\partial y} = f_y}$$



ex(1):

$$f(x, y) = x^2 + 3xy + y - 1$$

$$\frac{\partial f}{\partial x} = f_x = 2x + 3y$$

$$\frac{\partial f}{\partial y} = f_y = 0 + 3x + 1 = 3x + 1$$

ex(2)

$$f(x, y) = y \sin(xy)$$

$$\frac{\partial f}{\partial x} = f_x = y \cos(xy) = y^2 \cos(xy)$$

$$\begin{aligned} \frac{\partial f}{\partial y} = f_y &= y \times \cos(xy) \times x + \sin(xy) \\ &= yx \cos(xy) + \sin(xy) \end{aligned}$$

(ex3)

$$f(x, y) = \frac{0 - 2y}{y + \cos x}$$

$$f_x = \frac{\partial f}{\partial x} = \frac{0 + 2y \times -\sin x}{(y + \cos(x))^2} =$$

$$= \frac{-2y \sin x}{(y + \cos(x))^2}$$



$$f_y = \frac{\partial F}{\partial y} = \frac{(y + \cos x) + 2 - 2y + 1}{(y + \cos(x))^2}$$

$$= \frac{2y + 2\cos x - 2y}{(y + \cos x)^2} = \frac{2\cos x}{(y + \cos x)^2}$$

ex(4)

$$f(x, y) = \cos\left(\frac{4}{x}\right) e^{x^2y - 5y^3}$$

$$f_x = \frac{\partial f}{\partial x} = \cos\left(\frac{4}{x}\right) \cdot e^{x^2y - 5y^3} \cdot 2xy$$

$$+ e^{x^2y - 5y^3} \cdot \sin\left(\frac{4}{x}\right) \cdot \frac{4}{x^2}$$

$$= 2xy \cos\left(\frac{4}{x}\right) e^{x^2y - 5y^3} + \frac{4}{x^2} \sin\left(\frac{4}{x}\right) e^{x^2y - 5y^3}$$

$$f_y = \frac{\partial f}{\partial y} = \cos\left(\frac{4}{x}\right) e^{x^2y - 5y^3} \cdot (x^2 - 5 \cdot 3y^2)$$

$$= (x^2 - 15y^2) \cos\left(\frac{4}{x}\right) e^{x^2y - 5y^3}$$



Ex 5

$$P(x, y) = \sqrt{x^2 + \ln(5x - 3y^2)}$$

$$P_x = \frac{\partial P}{\partial x} = \frac{1 \times 2x + \frac{5}{5x - 3y^2}}{2\sqrt{x^2 + \ln(5x - 3y^2)}}$$

$$= \frac{2x + \frac{5}{5x - 3y^2}}{2\sqrt{x^2 + \ln(5x - 3y^2)}}$$

$$P_y = \frac{\partial P}{\partial y} = \frac{1 \times \left(0 + \frac{-3 \times 2y}{5x - 3y^2}\right)}{2\sqrt{x^2 + \ln(5x - 3y^2)}}$$

$$= \frac{\frac{-6xy}{5x - 3y^2}}{2\sqrt{x^2 + \ln(5x - 3y^2)}}$$

$$= \frac{-3xy}{(5x - 3y^2)\sqrt{x^2 + \ln(5x - 3y^2)}}$$



Ex 5: $P(x, y, z) = x \sin(y + 3z)$

(s.a.)

$$P_x = \frac{\partial P}{\partial x} = \sin(y + 3z)$$

$$P_y = \frac{\partial P}{\partial y} = x \cos(y + 3z)$$

$$= x \cos(y + 3z)$$

$$P_z = \frac{\partial P}{\partial z} = x \cos(y + 3z) \times 3$$

$$= 3x \cos(y + 3z)$$

///



(II) second-order partial Derivatives

$$\Rightarrow \frac{\partial^2 f}{\partial x \partial y} = f_{yx}$$

2 ← 1

→ متساوية بالانابة $x \leftarrow y$

$$\frac{\partial^2 f}{\partial x^2} = f_{xx} \quad , \quad \frac{\partial^2 f}{\partial y^2} = f_{yy}$$

$$\frac{\partial^2 f}{\partial x \partial y} = f_{yx} \quad , \quad \frac{\partial^2 f}{\partial y \partial x} = f_{xy}$$

لازم $\Rightarrow$ تكون متساوية المشتقة الأولى f_x and f_y

(Note)

لكانت الدالة $(f_{xy} = f_{yx}) \Rightarrow$

متصلة (Continuous)

#



ex ① :- Find All 2nd order derivatives :-

$$f(x, y) = x \cos y + y e^x \quad (-s=1)$$

$$\frac{\partial f}{\partial x} = \cos y + y e^x \rightarrow \textcircled{1}$$

$$f_y = \frac{\partial f}{\partial y} = -x \sin y + e^x \rightarrow \textcircled{2}$$

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} = 0 + y e^x = y e^x$$

$$f_{yy} = -x \cos y + 0 = -x \cos y$$

$$f_{xy} = -\sin y + e^x \rightarrow *$$

$$f_{yx} = -\sin y + e^x$$

((Note))

$$f_{xy} = f_{yx} = -\sin y + e^x$$

continuous



* Directional Derivative * الاشتقاق الاتجاهي

ان كانت الحالة $f(x, y, z)$ قابلة للاشتقاق عند النقطة (P) وكان $\vec{V}$ متجه في الفراغ ونقطة بداية P فإن المشتقة الاتجاهية للدالة (f) عند النقطة P وباتجاه المتجه $\vec{V}$ هي :

$$\Rightarrow D_{\vec{u}} f = \nabla f \cdot \vec{u}$$

$\hookrightarrow \vec{u} \rightarrow$ unit vector

$\hookrightarrow \nabla f =$ gradient vector at (P)

I) Vector (V)

in Direction of f

$\Rightarrow$ any point $x \in \mathbb{R}^n$

$$\mathbb{R}^2 \Rightarrow V = \square i + \square j$$

ex: $(2, 3) \in \mathbb{R}^2 \Rightarrow V = 2i + 3j$

$$\mathbb{R}^3 \Rightarrow V = \square i + \square j + \square k$$

ex:

$$(-1, 4, -5) \in \mathbb{R}^3 \Rightarrow V = -i + 4j - 5k$$



I) unit vector (متجه وحدة)

$$\vec{u} = \frac{\vec{V}}{\|\vec{V}\|}$$

For $\mathbb{R}^2$

$$\vec{u} = \frac{x}{\|\vec{V}\|} \vec{i} + \frac{y}{\|\vec{V}\|} \vec{j}$$

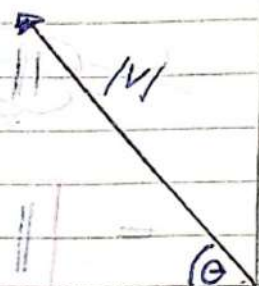
$$\vec{u} = \cos \theta \vec{i} + \sin \theta \vec{j}$$

(a)

(b)

x

y



For $\mathbb{R}^3$

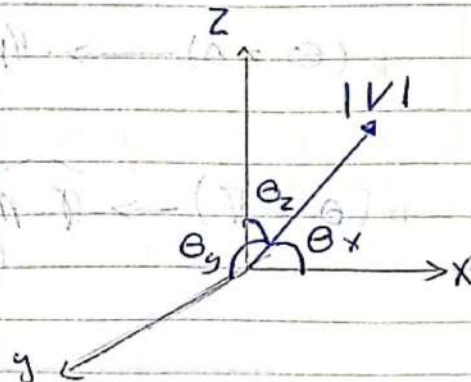
$$\vec{u} = \frac{x}{\|\vec{V}\|} \vec{i} + \frac{y}{\|\vec{V}\|} \vec{j} + \frac{z}{\|\vec{V}\|} \vec{k}$$

$$\vec{u} = \cos \theta_x \vec{i} + \cos \theta_y \vec{j} + \cos \theta_z \vec{k}$$

(a)

(b)

(c)



III) $\nabla f(x, y, z) \Rightarrow$ Gradient of function

$$\nabla f = F_x \vec{i} + F_y \vec{j} + F_z \vec{k}$$

$$\nabla f = \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial z} \vec{k}$$



Note

① $D_{\vec{u}} f \Rightarrow$ Directional derivative is Value
 في اتجاه (نا) من حيث هـ

* ②

$$D_{\vec{u}} f = \nabla f \cdot \vec{u} = \|\nabla f\| \|\vec{u}\| \cos \theta$$

~~$$= \|\nabla f\| \cos \theta$$~~

$$= \|\nabla f\| \cos \theta$$

$(\theta = 0) \rightarrow$ (maximum value)
 ($\cos \theta = 1$)

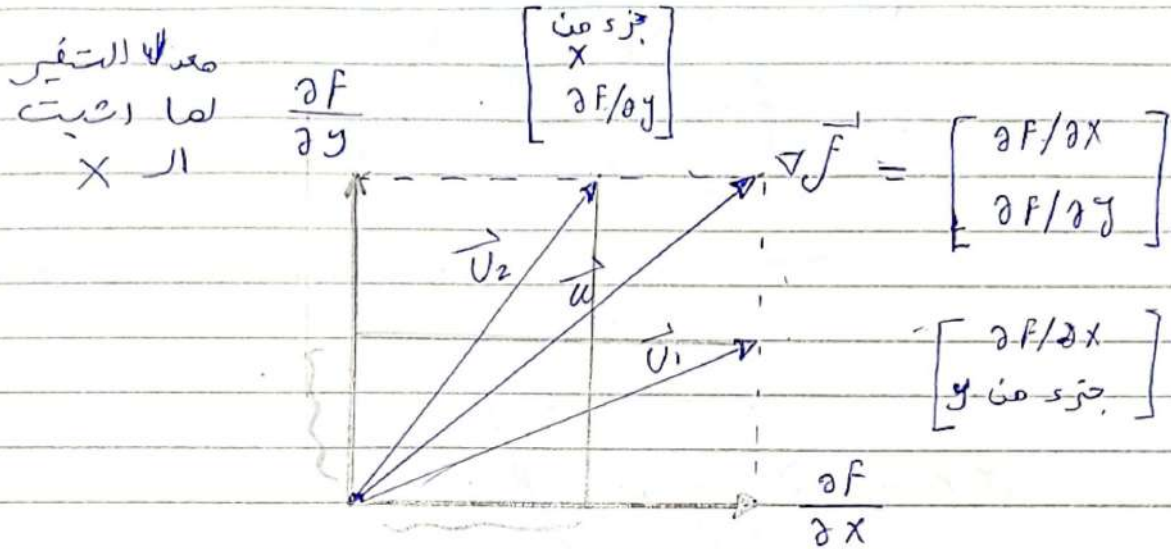
$(\theta = \pi) \rightarrow$ (minimum value)
 ($\cos \theta = -1$)

#

$$D_{\vec{u}} F = f_x a + f_y b + f_z c$$



* Gradient Vector's Intuition:



معدل التغير لما اصبحت الـ y

1. اتجاه موجهة العدد التغير في الـ x و جزء من y : $\vec{U}_1$

$\vec{U}_2$: $\vec{U}_1$ في الـ y و جزء من x

$\vec{U}$: $\vec{U}_1$ و $\vec{U}_2$ معاً



* $\vec{u}$ الاتجاه هو الاتجاه الأكبر ميلًا :

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}$$

لذلك : اتجاه الـ (gradient vector) $\vec{u}$ في اتجاه الـ (steepest ascent) اتجاه الـ $\vec{u}$

$$\nabla f \cdot \vec{u} = \|\nabla f\| \|\vec{u}\| \cos \theta \quad (\cos \theta = 1)$$

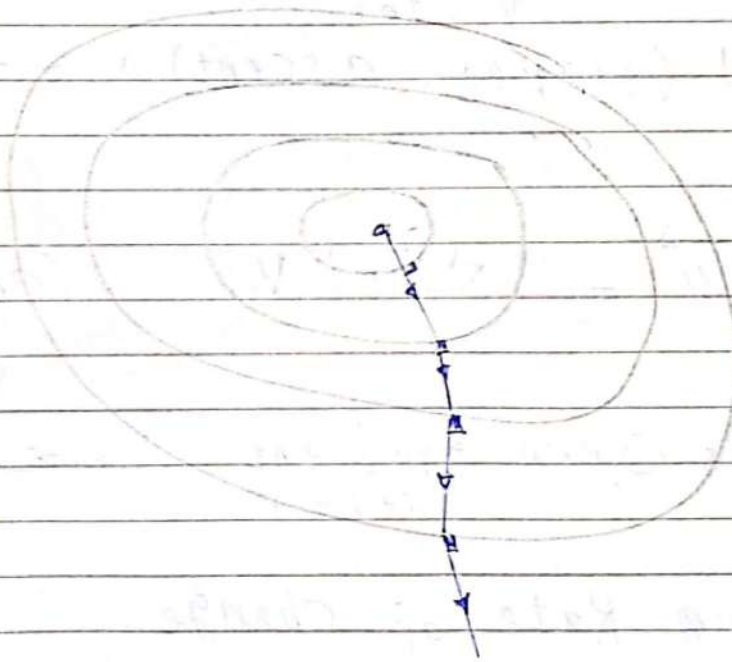
لـ $\vec{u}$ اتجاه الـ gradient vector ينطبق على اتجاه الـ $\vec{u}$ $\theta = 0$

→ Maximum Rate of change.

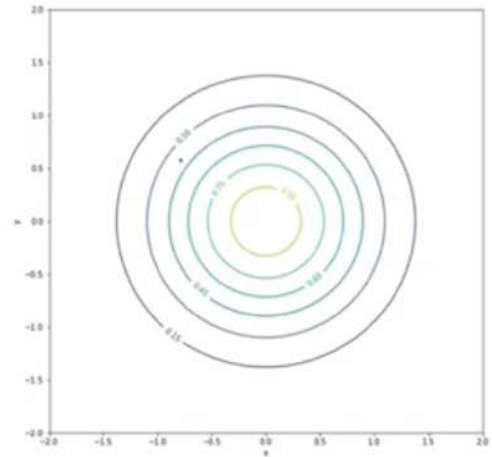
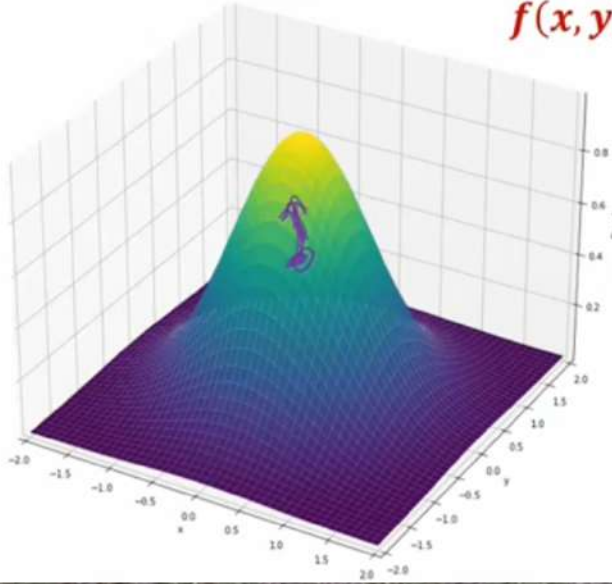


ملاحظة

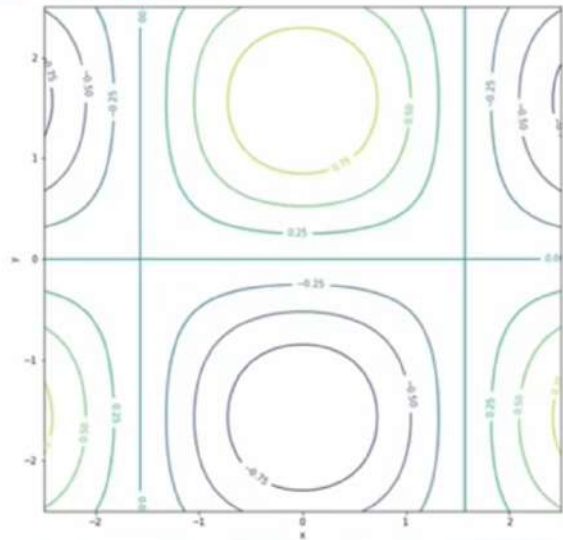
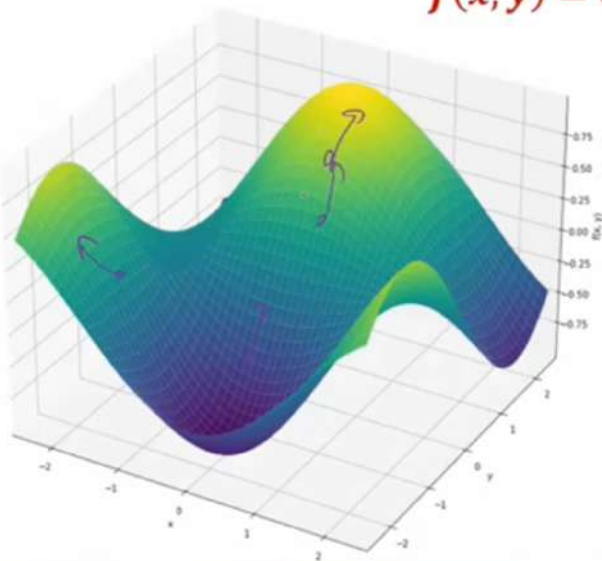
Contour Plot $\leftarrow$ gradient Vector $\leftarrow$ 1



$$f(x, y) = e^{-(x^2+y^2)}$$



$$f(x, y) = \cos(x) \sin(y)$$



* Gradient Vector Field:

→ is the gradient vector evaluated at all points in its Domain.

$$\text{ex: } \nabla f(x, y) = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

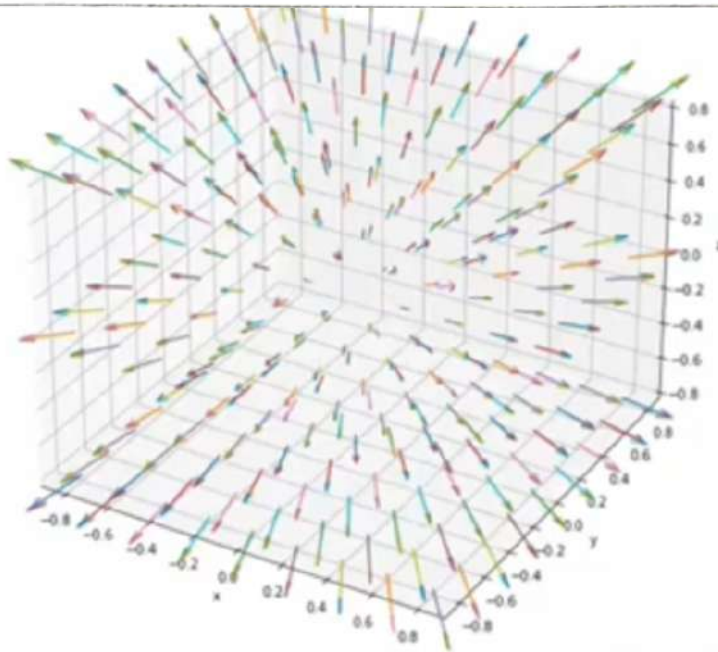
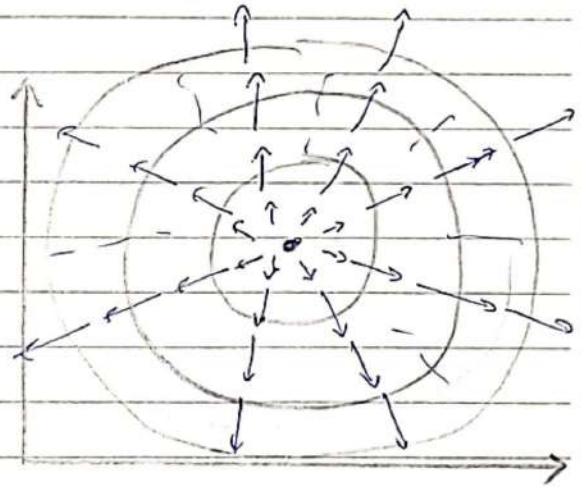
$$\nabla f(0, 0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

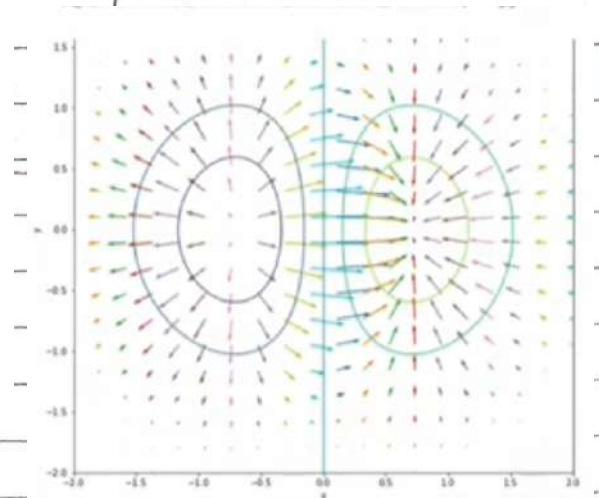
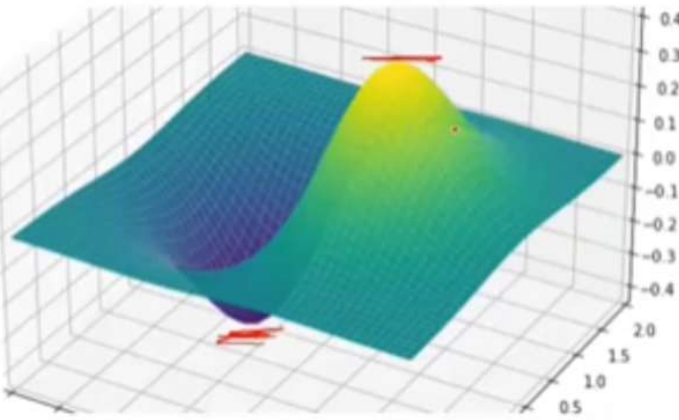
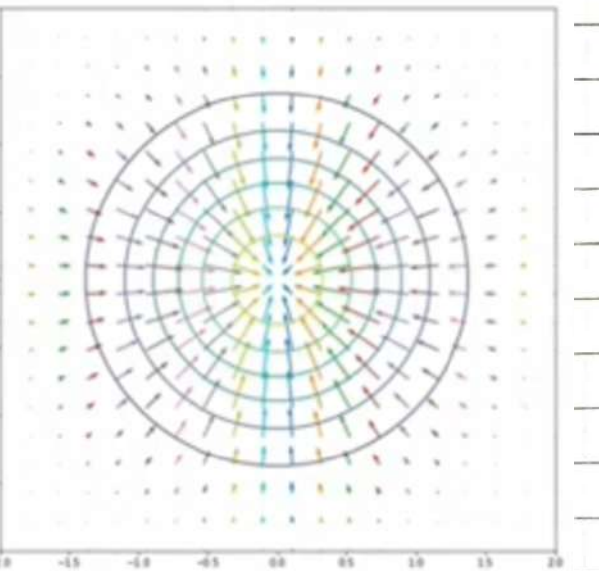
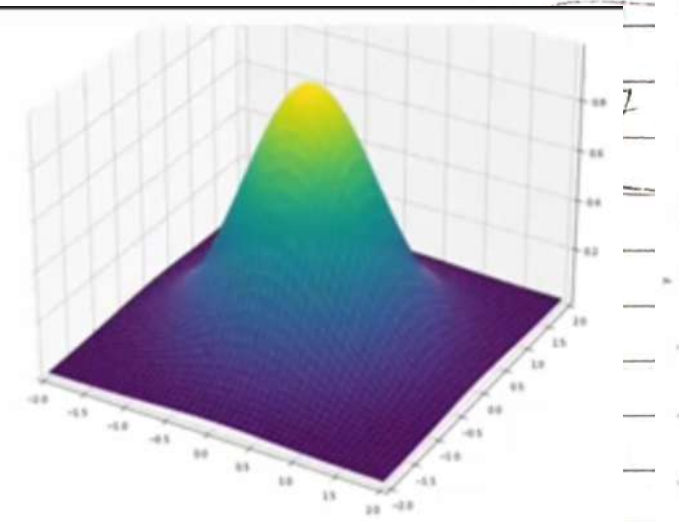
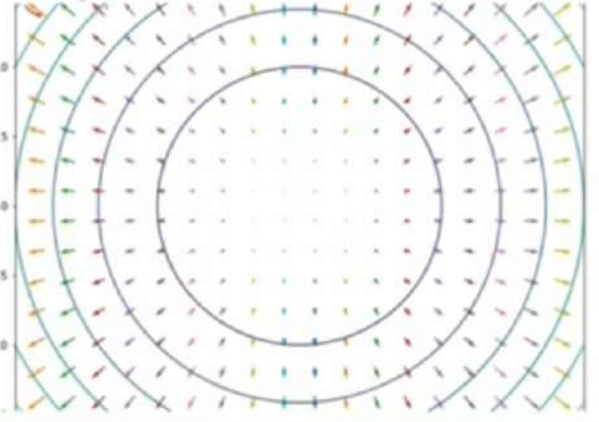
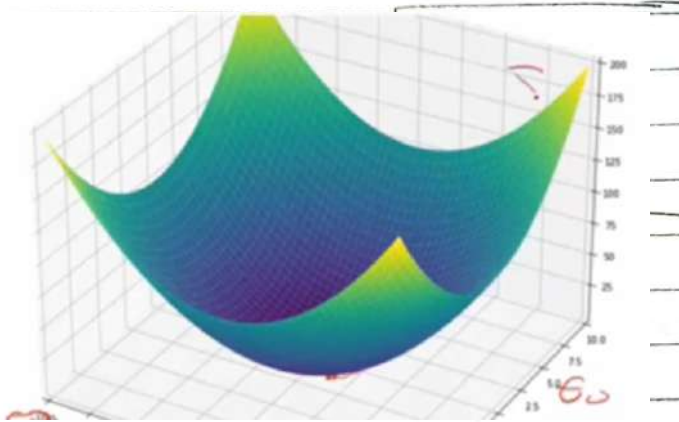
$$\nabla f(1, 0) = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\nabla f(0, 1) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

}

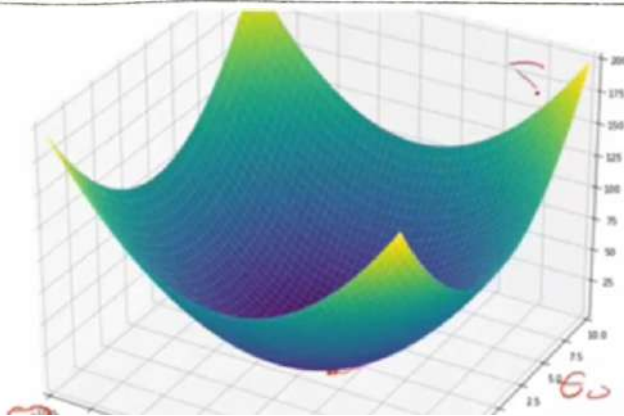
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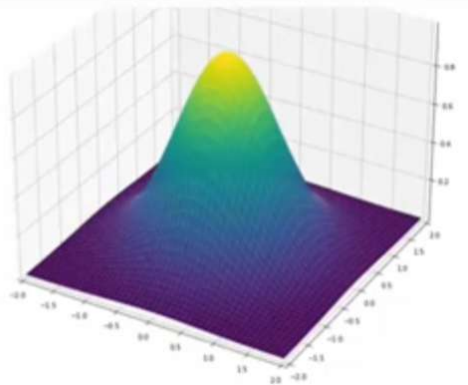


$$\ast \left\{ \|\nabla f(x, y)\| = 0 \right\}$$

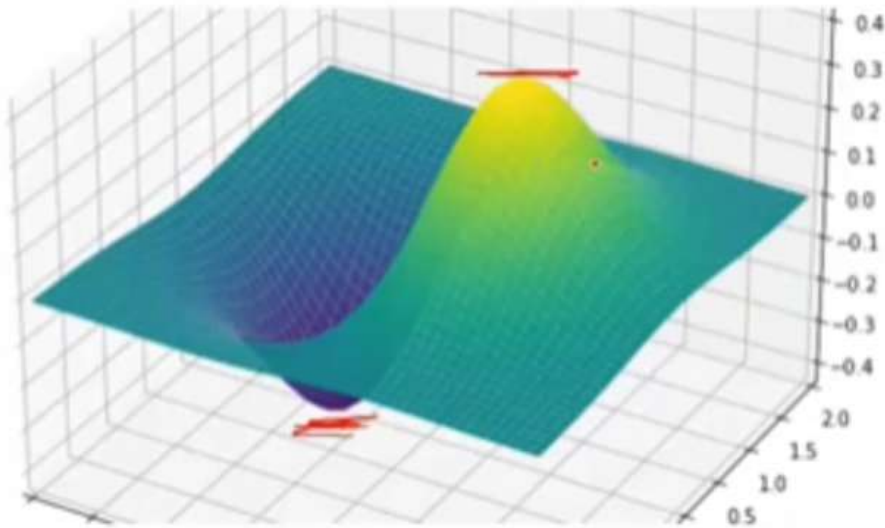
Case (I) : Minimum



Case (II) : Maximum :



Case (III): saddle point



الملاحظة

لما نحن العساسة الى نموذج كذا ال Training
 ال ال model ال parameters كذا
 ال (saddle point)

ال error كذا ال هو ال min



* Hessian Matrix:

→ is a square matrix of second-order partial derivatives of a scalar-valued func.

$$f(x_1, x_2, \dots, x_n)$$

$$H(f) = \begin{vmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{vmatrix}$$

→ tells about the Curvature of multivariable function.



if : $H(f)$:

Positive
Definite
Matrix



Minimum

Convex

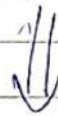
Negative
Definite
Matrix



Maximum

Concave

Indefinite
Matrix



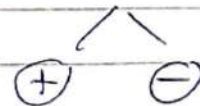
Saddle point

Test

All $\lambda_i > 0$

All $\lambda_i < 0$

Mixed signs
of λ_i



ex:

$$f(x, y) = \cos(x) \sin(y)$$

(سا ١)

$$\nabla f(x, y) = \begin{bmatrix} \partial f / \partial x \\ \partial f / \partial y \end{bmatrix} = \begin{bmatrix} -\sin(x) \sin(y) \\ \cos(x) \cos(y) \end{bmatrix}$$

$$Hf(x, y) = \begin{bmatrix} -\cos(x) \sin(y) & -\sin(x) \cos(y) \\ -\sin(x) \cos(y) & -\cos(x) \sin(y) \end{bmatrix}$$

#



II) Multi Variable chain Rule;

$$\frac{d}{dt} f(\vec{V}(t)) = \nabla f(\vec{V}(t)) \cdot \vec{V}'(t)$$

$$f(x, y, z) = \sin(x) e^{yz^2}$$

$$x(t) = t - 1 \quad \rightarrow \quad dx/dt = 1$$

$$y(t) = t^2 \quad \rightarrow \quad dy/dt = 2t$$

$$z(t) = 1/t \quad \rightarrow \quad dz/dt = -1/t^2$$

$$\frac{\partial f}{\partial x} = \cos(x) e^{yz^2}$$

$$\frac{\partial f}{\partial y} = z^2 \sin(x) e^{yz^2}$$

$$\frac{\partial f}{\partial z} = 2zy \sin(x) e^{yz^2}$$

$$\frac{\partial f}{\partial t} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial t} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial t}$$



Note

$$\nabla f(x, y, z) = \begin{bmatrix} \partial f / \partial x \\ \partial f / \partial y \\ \partial f / \partial z \end{bmatrix}$$

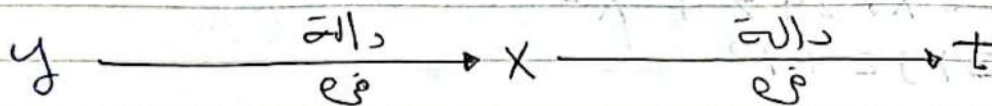
$$\vec{J} = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix} \longrightarrow \vec{V} = \begin{bmatrix} \partial x / \partial t \\ \partial y / \partial t \\ \partial z / \partial t \end{bmatrix}$$

$$\frac{\partial f}{\partial t} = \nabla f \cdot \vec{V}(t)$$



* Chain Rule * قاعده السلسلة

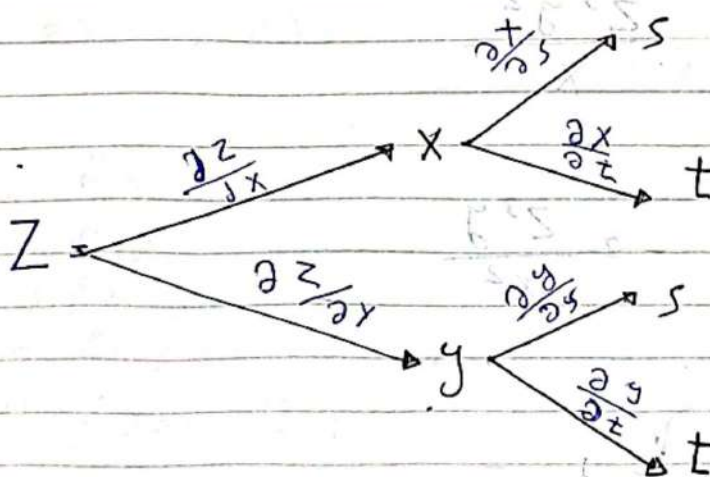
$$(y = f(x)) \text{ و } (x = g(t))$$



$$\Rightarrow \frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

* The tree diagram *

$$Z = f(x, y), \quad x = g(s, t), \quad y = h(s, t)$$



$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}$$

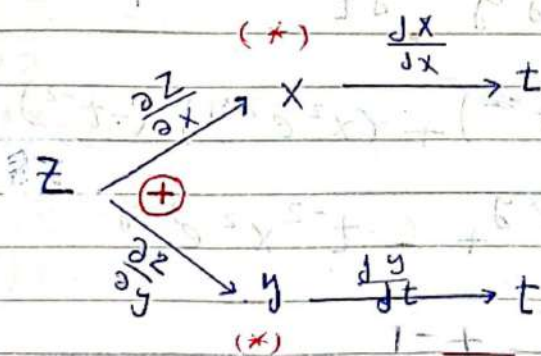
(ملاحظة)

① لمشتقات (الفرع الواحد) ← $\boxed{\times}$ (الضرب)

② لفرع آخر نفس الاشتقاق ← $\boxed{+}$ (الجمع)

③ اشتقاق بالنسبة لمتغير مستقل (بوحده) $\leftarrow (x)$ (d)

④ اشتقاق بالنسبة لمتغير جزئي (مع متغير ثانية) $\leftarrow (x, y, z)$ (∂)

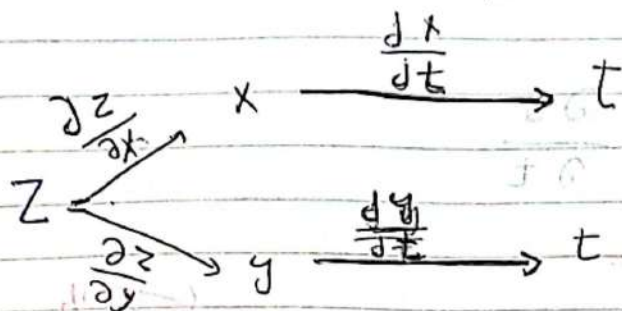


$$\left(\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt} \right)$$



EX:-

$$Z = X e^{xy}, \quad X = t^2, \quad y = t^{-1} \quad (s.a)$$



$$\frac{\partial Z}{\partial X} = X \cdot y e^{xy} + e^{xy} \cdot 1$$

$$\frac{\partial Z}{\partial Y} = X^2 e^{xy}$$

$$\frac{dX}{dt} = 2t, \quad \frac{dY}{dt} = -t^{-2}$$

$$\frac{dZ}{dt} = \frac{\partial Z}{\partial X} \frac{dX}{dt} + \frac{\partial Z}{\partial Y} \frac{dY}{dt}$$

$$= (X y e^{xy} + e^{xy}) (2t) + (X^2 e^{xy}) (-t^{-2})$$

$$= 2 X y t e^{xy} + 2 t e^{xy} - t^{-2} X^2 e^{xy}$$

$$((\text{Note})) \Rightarrow X = t^2, \quad y = t^{-1}$$

$$\therefore = 2 \cdot t^2 \cdot t^{-1} \cdot e^t + 2 t e^t - t^{-2} \cdot t^4 e^t$$

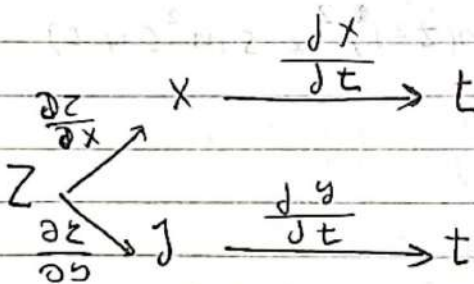
$$= 2 t^2 e^t + 2 t e^t - t^2 e^t$$



$$= t^2 e^t + 2 t e^t$$

$$\frac{dz}{dt} = t^2 e^t + 2 t e^t$$

$$* Z = x^2 y^3 + y \cos x, \quad x = \ln(t^2), \quad y = \sin(4t)$$



$$\frac{\partial Z}{\partial x} = 2 y^3 x + - y \cdot \sin x$$

$$= 2 y^3 x - y \sin x$$

$$\frac{\partial Z}{\partial y} = 3 x^2 y^2 + \cos x$$

$$\frac{dx}{dt} = \frac{2t}{t^2} = 2t^{-1}$$

$$\frac{dy}{dt} = 4 \cos(4t)$$

$$\therefore \frac{dz}{dt} = \frac{\partial Z}{\partial x} \frac{dx}{dt} + \frac{\partial Z}{\partial y} \frac{dy}{dt}$$



$$= (2y^3x - \sin x + y)(2t^{-1}) + (2x^2y^2 + \cos x)$$

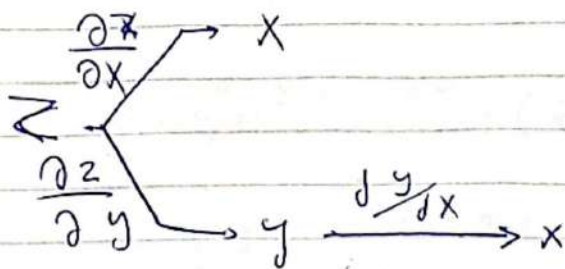
$$= (2 \times (\sin(4t))^3 \times \ln(t^2))(2t^{-1})$$

$$+ (2 \times (\ln(t^2))^2 \times (\sin(4t))^2 + \cos(\ln t^2))(4t \cos 4t \times \ln t^3)$$

$$= 8 \sin^3(4t) \times \frac{\ln t^2}{2t} + 4 (\ln(t^2))^2 \times \sin^2(4t) + 4 \cos(\ln t^2) \cos 4t (\ln t)^3$$

ex(3): Find $\frac{\partial Z}{\partial x}$ for

$$Z = x \ln(xy) + y^3 \quad y = \cos(x^2 + 1)$$



$$\frac{\partial Z}{\partial x} = x \times \frac{y}{xy} + \ln(xy) + 0$$

$$= 1 + \ln(xy)$$



* Jacobian matrix :

→ is a natural generalization of Vector valued Functions of several variables of the derivative & the differential of the usual functions.

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$J_f = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \dots & \frac{\partial f}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \nabla^T f_1 \\ \vdots \\ \nabla^T f_m \end{bmatrix}$$

$$J_f(x) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

$m \times n$



$$f(x_1, x_2) = f(\bar{x}) = x_1^2 - x_2^2$$

$$x_1(u_1, u_2) = 2u_1 + 3u_2$$

$$x_2(u_1, u_2) = 2u_1 - 3u_2$$

$$X(u) = \begin{bmatrix} x_1(u_1, u_2) \\ x_2(u_1, u_2) \end{bmatrix}$$

$$u_1(t) = \cos(t/2)$$

$$u_2(t) = \sin(2t)$$

$$u(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix} = \begin{bmatrix} \cos(t/2) \\ \sin(2t) \end{bmatrix}$$



$$F(X(u(t)))$$

$$F' = \frac{dF}{dt} = \frac{dF}{dX} \cdot \frac{dX}{du} \cdot \frac{du}{dt}$$

$$X \frac{dF}{dX} = \begin{bmatrix} dF/dX_1 \\ dF/dX_2 \end{bmatrix} = \begin{bmatrix} 2X_1 \\ -2X_2 \end{bmatrix} \quad \checkmark$$

$$X \frac{du}{dt} = \begin{bmatrix} du/dt_1 \\ du/dt_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \sin(t/2) \\ 2 \cos(2t) \end{bmatrix} \quad \checkmark$$

$$X \frac{dX}{du} = X \begin{bmatrix} \frac{\partial X_1}{\partial u_1} & \frac{\partial X_1}{\partial u_2} \\ \frac{\partial X_2}{\partial u_1} & \frac{\partial X_2}{\partial u_2} \end{bmatrix} \quad \checkmark$$

↪ Jacobian matrix



* Numerical solution:

Intro:

$$y = P(x) = x^2 + 2x - 1$$

$$a = 1$$

$$b = 2$$

$$c = -1$$

$$x = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\left. \begin{array}{l} \rightarrow x = 0.5 \\ x = -3 \end{array} \right\} \begin{array}{l} \text{the perfect solution} \\ \text{accuracy} = 100\% \end{array}$$

⇒ Analytical solution

* لها درجة الـ Polynomial تزيد عن (2)

يبقى من الصعب الحل بشكل Analytical

Numerical
solution

لذلك بدرجة لا



Steps of Numerical solution :-

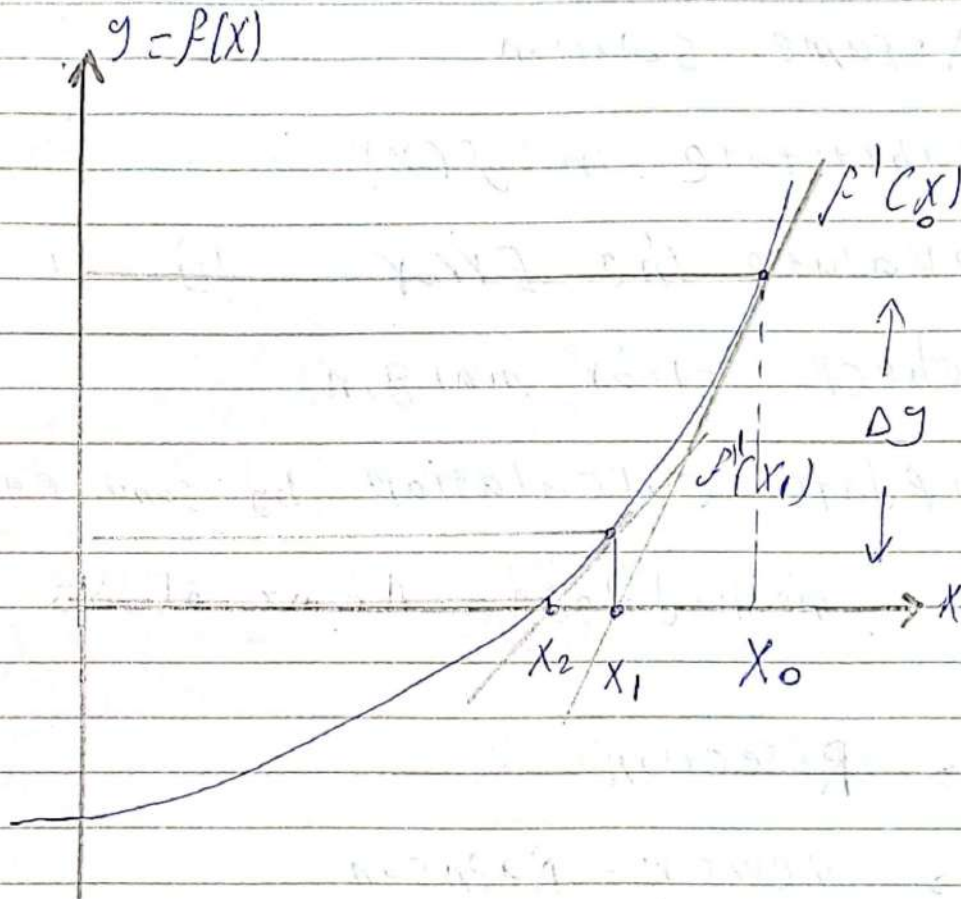
- ① Assume solution
- ② substitute in $P(X)$
- ③ evaluate the Error $|\hat{y} - y|$
- ④ check error margin
- ⑤ update calculation by some equations

مختلف حسب methodology

- Bisection
- Newton - Raphson
- secant
- Forward different
- Central



* Newton-Raphson method



الـ update eq. من أجل gradient

$$f'(x_0) = \frac{f(x_0) - 0}{x_0 - x_1}$$

$$\therefore x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$



Form:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Steps

① Start with an initial guess x_0 .

② Compute the next Approximation

using:
$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

③ Stop when $|x_{n+1} - x_n| \Rightarrow$ very small
(converged)



Ex: Det. the solution Approximated to
 $f(x) = \cos(x) - x$
lies in the interval $[0, 2]$.

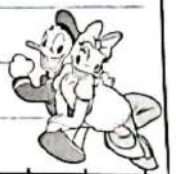
رسم

① let: $x_0 = 1 \in [0, 2]$

② $f(x) = \cos(x) - x = 0$

$$f'(x) = -\sin(x) - 1$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$



(3)

n	x_n	$f(x_n)$	$f'(x_n)$
0	1	-0.459697	-1.84147
1	0.750363	-0.01892	-1.681904
2	0.739112	-0.000004	-1.67363
3	0.739085	-3.0939	-1.6736
4	0.739085		
5	✓		
6			
7			

$$\therefore x_{\text{solution}} = 0.739085$$



* limitations :-

(*) معتمدة على $f'(x)$

(a) لا يمكن تكونه (undefined)

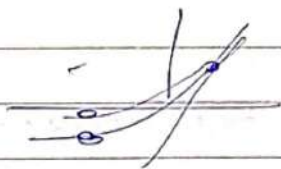
$$f'(x) = \infty$$

(b) لا يكون $f'(x) = 0$

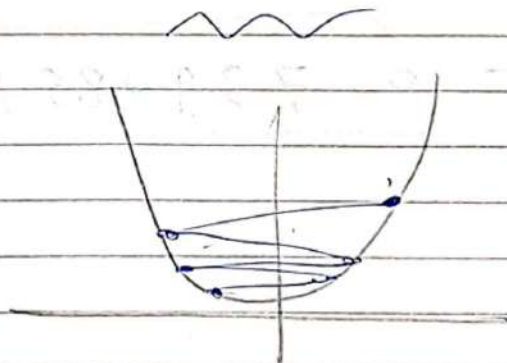
(c) لا فرقته صغيرة جداً

==> فيحصل update كبير جداً ↑

فيحصل overshoot



ويحصل (oscillation) حوالته الجذر بدون استقرار



* Taylor Series Approximation :

→ is a way to Approximate a complicated function using a polynomial built from the function's derivatives at a single point.

⇒ used to Reconstruct the function everywhere else.

Form

Condition : $f(x)$ is infinitely differentiable at a point (a) .

then :

No: Approx.

1st order approx.

$$f(x) \approx \boxed{f(a)} + \boxed{f'(a)(x-a)}$$

$$+ \boxed{\frac{f''(a)}{2!}(x-a)^2} + \frac{f^{(3)}(a)}{3!}(x-a)^3 + \dots$$

2nd order Approx.

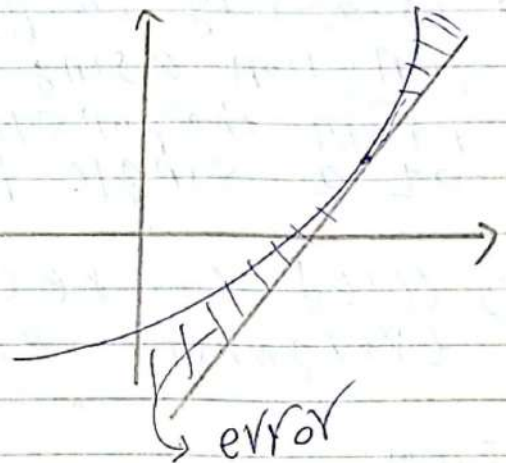


كل نقطة: كل ما أزدور ال Approx.
كل ما أقرب إلى ال Curve الحقيقي

* proof of Taylor series using Newton method

$$f'(s) = \frac{f(s) - f(x)}{s - x}$$

$$s - x = \frac{f(s) - f(x)}{f'(s)}$$



$$(s - x) f'(s) + f(x) = f(s)$$

$$f(x) = f(s) + f'(s)(s - x)$$

1st order Approximation

خط التقريب الأول (1st order Approximation) خط التقريب الأول (1st order Approximation)
خط التقريب الأول (1st order Approximation) خط التقريب الأول (1st order Approximation)

St-P

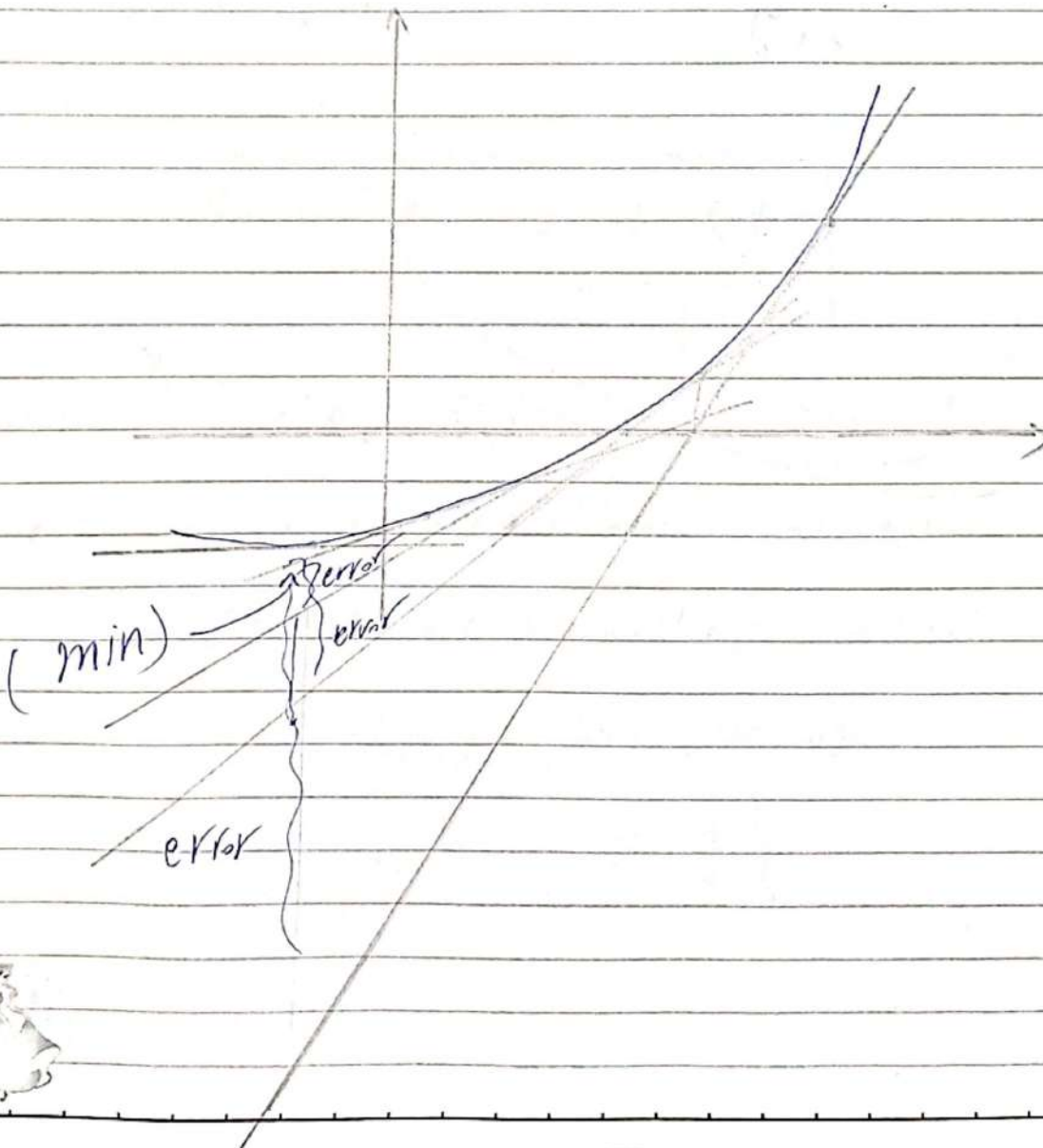


في الـ optimization انا هدف في الوصول لـ min

صفت لي جاد الـ (Roots)

لذلك ما نشان اوصول لـ min باسـ time

لنر : ا في steps اسي
 ج 1 step = تاوت اسي

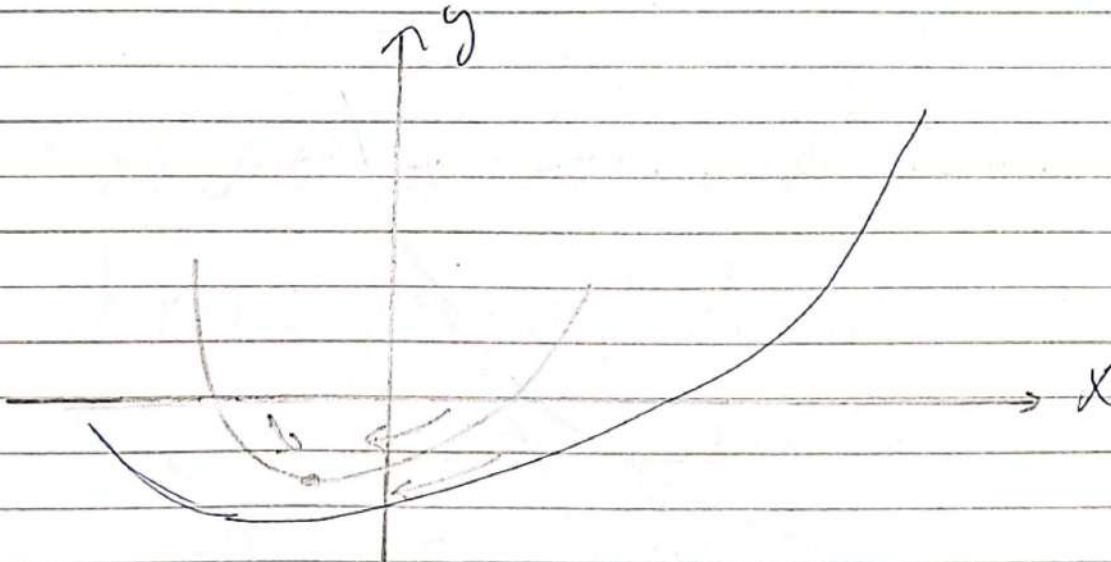


الجزء الثاني :

نصل Approx. بالاضافة Curve

① f في $step$ واحدة

② $\min$ curve هو $\min$ تقريباً الأعلى



* Exponentially Weighted Moving Average :- (EWMA)

→ EWMA

→ is a statistical technique used to smooth time series data by giving more weight to recent observations while not completely discarding older data.

Form :

$$S_t = \alpha X_t + (1 - \alpha) S_{t-1}$$

S_t : smoothed value at time t

X_t : actual observation at time t

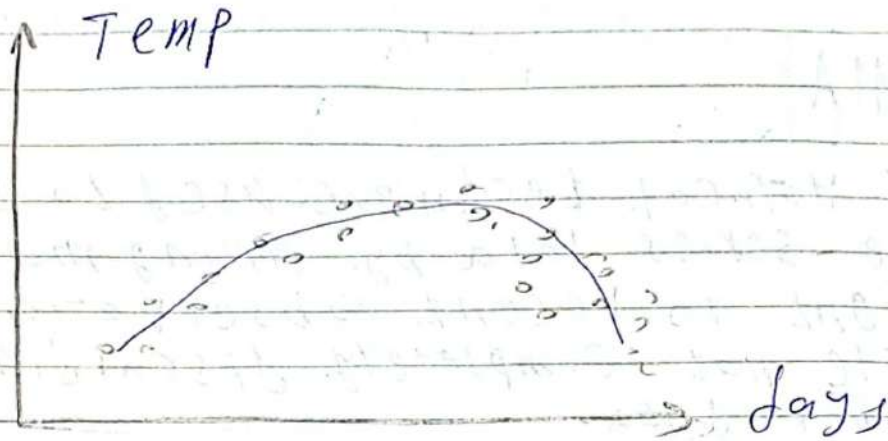
S_{t-1} : previous smoothed value

α : smoothing parameter

$$0 < \alpha \leq 1$$

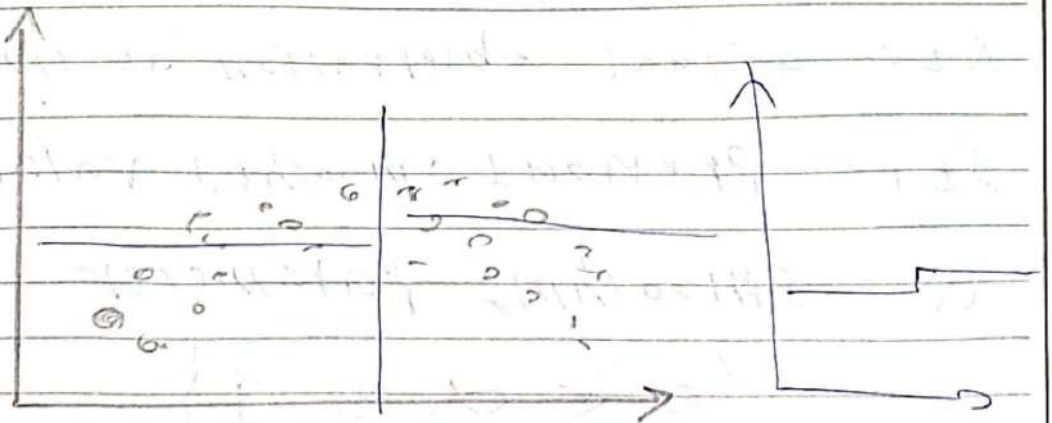


* Intuition of EWMA



عائز أوجد له curve الى بقدر
 في model كذا data بدون ما يدخل
 (overfitting)

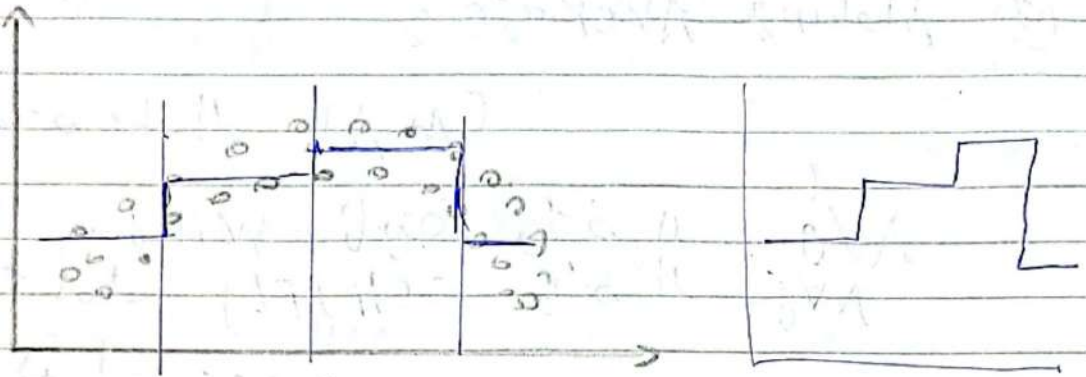
(1)



× هقسم لنصفين ما جيب لا avg
 لكل نصف
 ← error كبير X X



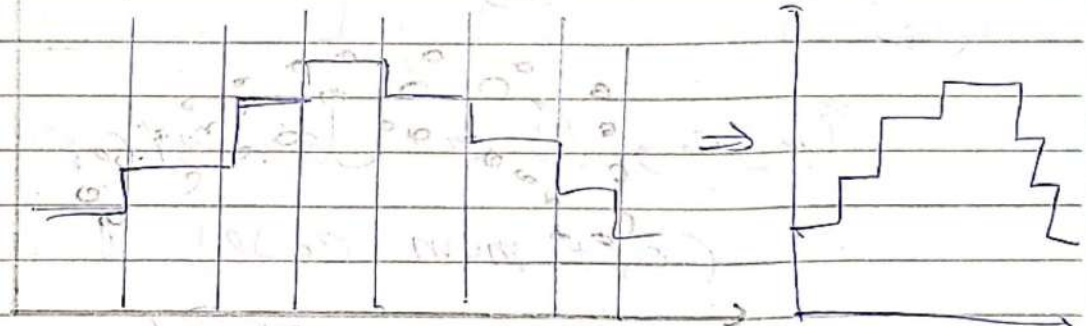
②



→ هقسم لعدة فترات وأخذ الـ avg لكل فترة
 → الـ $error$ هيقول وكنيت مازال كبير

③

كل الـ avg (10 days) هأخذ الـ (AVG)



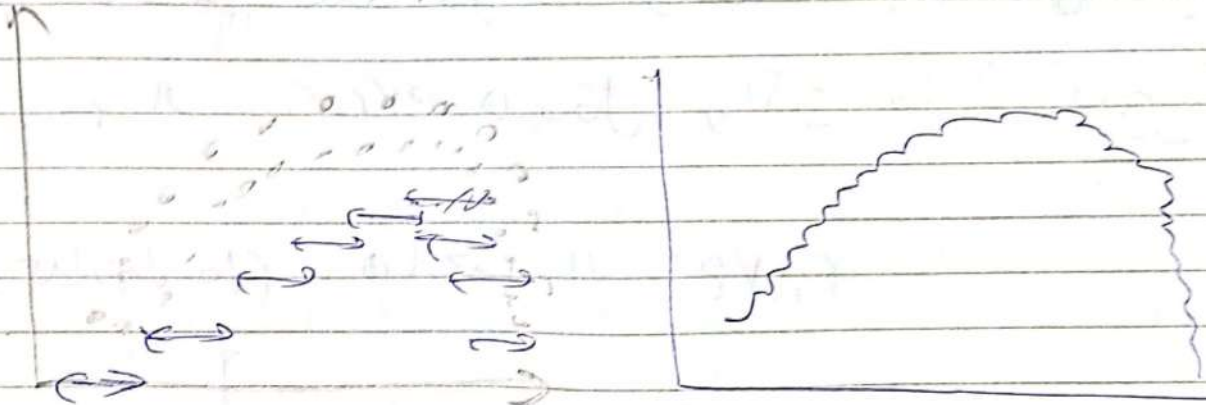
→ الـ $error$ هيقول أستر وكنيت مازال قليل



(4) Moving Average :

هناك نوعان من (MA)

1- Window ثابتة وأخذ AVG
 2- متحرك (shift) وأخذ AVG
 وهكذا ...



هنا $error$ قليل جداً وتقريباً

وصلت إلى (optimum case)

ولكن هناك مشكلة

لـ (a) تحتاج data كثير جداً

لـ (c) تحتاج إلى Iterations كثير جداً

Computationally
expensive



EWMA

ننظر بالتحديد

لي

فيها (recursive method)

يعني ما حسب value واحدة فقط

ولكنها تخزن (information) عند كل الى قبلها

$$V_t = B V_{t-1} + (1-B) T_t$$

 $B = 0.9 \rightarrow$ history على ال

 $B = 0.1 \rightarrow$ current على ال


$$V_0 = 0$$

$$V_1 = (1-B) T_1$$

$$V_2 = B V_1 + (1-B) T_2$$

$$V_3 = B(B(1-B) T_1 + (1-B) T_2) + (1-B) T_3$$

$$V_3 = B^2(1-B) T_1 + B(1-B) T_2 + (1-B) T_3$$

$$V_4 = B V_3 + (1-B) T_4$$

$$= B^3(1-B) T_1 + B^2(1-B) T_2 + B(1-B) T_3 + (1-B) T_4$$

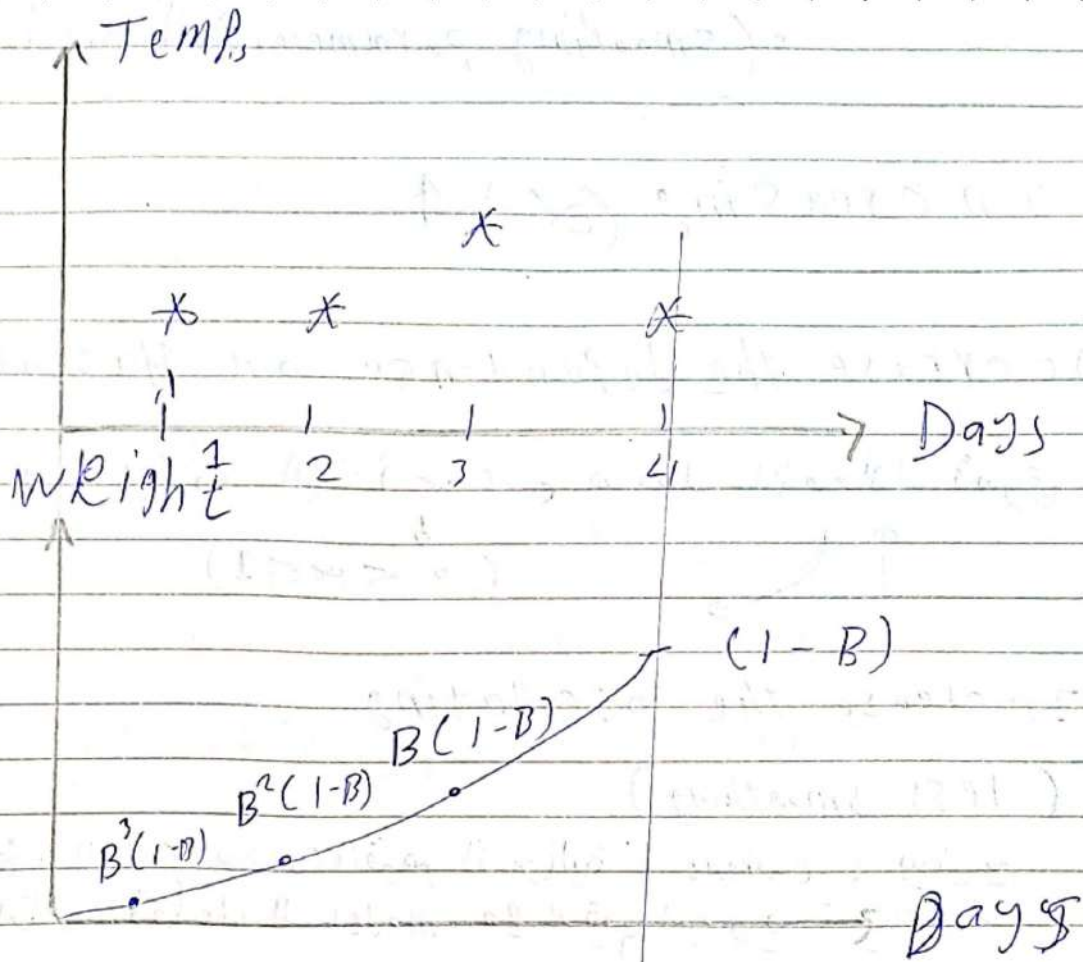
نلاحظ

كل ما أبعد عن النقطة

كل ما أهميتها فنقل

كل ما weight يتناقص فنقل





* تأثير الـ (smoothing parameter) :

① Increasing $(\alpha) \uparrow$

* Decrease the dependence on History $\downarrow$

لأنه كلما زادت (α) كلما قل اعتمادنا على التاريخ

$(0 < \alpha \leq 1)$



* Increase the oscillating

(less smoothing)

لأنه كلما زادت (α) كلما قل اعتمادنا على التاريخ (current) فيزيء
بالتالي قفنا على الـ model مع القيمة الجديدة بزيادة

* Fewer No. of iterations

لأنه كلما زادت (α) كلما قل الاعتماد على التاريخ decay أسرع

بالتالي هو يصل لـ stable state أسرع
أو بعد عدد iterations أقل



② decreasing (α) ↓

* increase the dependence on History

لأنه فقامت الـ $\downarrow (\alpha)$ يعطى أهمية أكبر لـ history

* Decrease the oscillating ((more smoothing))

لأنه عند تقليل الـ (α) يعطى وزناً أقل للتغيرات الصغيرة

"less sensitivity"

* many more iterations ::

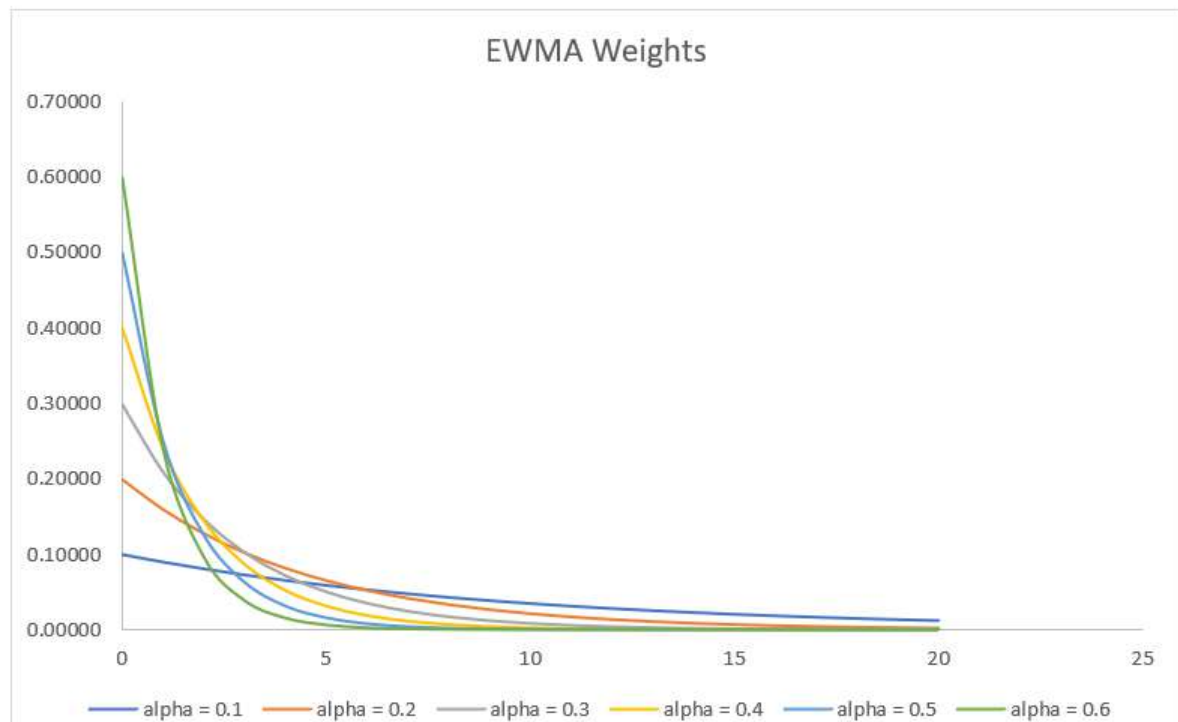
لأنه يتطلب عدد كبير من الـ iterations



* Comparison:

Exponentially Weighted Moving Average (EWMA) – Effect of α (Alpha)

Feature / Criterion	Increasing α (e.g., 0.1 $\rightarrow$ 0.5)	Decreasing α (e.g., 0.3 $\rightarrow$ 0.05)
Dependence on History	Decreases $\rightarrow$ Short Memory	Increases $\rightarrow$ Long Memory
Smoothing vs. Oscillating	More Oscillating (Noisier, tracks recent data closely)	More Smoothing (Less noisy, filters out short-term fluctuations)
Number of Iterations to Stabilize	Fewer $\rightarrow$ Converges faster, "forgets" its initial value quickly	More $\rightarrow$ Converges slower, requires a longer warm-up period
Sensitivity to New Data	High $\rightarrow$ Reacts strongly to new information	Low $\rightarrow$ Reacts slowly to new information
Effective Lookback Period	Shorter (similar to ~ 12 -period SMA)	Longer (similar to ~ 50 -period SMA)

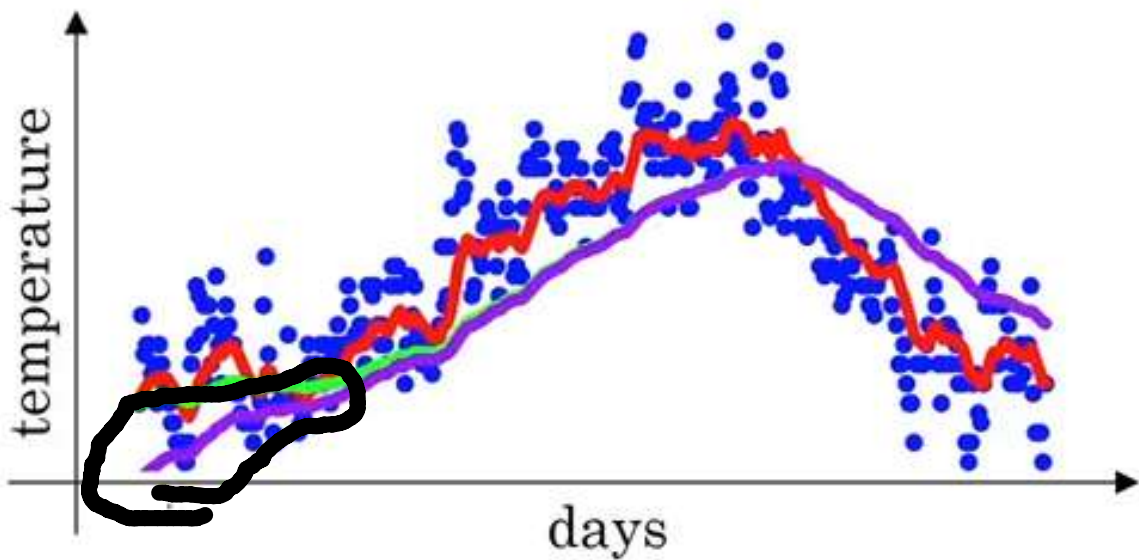


* Bias Correction :

لو فر حنا انه $\alpha = 0.98$

اننا محتاج أخذ الـ AV لأول 50 values على الأقل

لكنه الى بيحصل انه القيم اللي في البداية يكون ليها وزن أعلى وبالتالي بيحصل (Bias)



* solution of Bias ??

↳

$$\hat{V}_t = \frac{V_t}{1 - \beta^t}$$

$$V_1 = 0.02 T_1$$

$$\hat{V}_1 = \frac{0.02 T_1}{1 - (0.98)^1} = \checkmark$$

$$V_2 = (0.98)(0.02) T_1 + (0.02) T_2$$

$$\hat{V}_2 = \frac{\boxed{}}{1 - (0.98)^2} = \checkmark$$

⋮

